# Code generation from scratch

HW2 experiment report - work in progress
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## Current result

Earlier small-model checkpoints reached 180,000 updates; the selected 42.48M checkpoint is at step 102,563. Ten initial checkpoints, eight first-continuation checkpoints and 6 longer-continuation checkpoints were saved in Google Drive and scored by the course bot. The best public result is **62.19% (597/960)**. Latest verified rank is 9 of 16 at 2026-10-17T05:40:00+03:00. Its exact scored output is selected-scale-v1-102563.jsonl. This is the current best result across our training and decoding experiments and uses strict greedy decoding. Later sections distinguish historical settings. T4 timing remains deferred. No private result is available.

## Task and rules

Train an encoder-decoder Transformer from random weights using only the supplied SlavaYus/tinysms_qa_12m training split. Generate one Python program per test question. All 1,919 programs must be generated within 180 seconds on one NVIDIA T4. No pretrained weights, outside examples, model distillation, or agent-written target programs are used.

## Data and architecture

The allowed dataset contains 11,838,003 question/code pairs in 60 training files. The test file contains questions only. A byte-level BPE tokenizer with 8,192 tokens was fitted on shard 0 after reserving its first 1,024 rows. The initial baseline has 9,967,360 parameters: four encoder layers, four decoder layers, width 256, feed-forward width 768, four attention heads, gated GELU and dropout 0.1. It uses the T5 architecture with randomly initialized weights. The current selected model has 42,481,152 parameters, six encoder layers, four decoder layers, width 512 and feed-forward width 1536. It was trained from fresh random weights with the corrected full-coverage stream; its first complete pass ended at 92,456 updates. The separate historical runs and later second-pass continuation are recorded below.

The current recipe uses AdamW, BF16, batch 128, length grouping, prefetch and token-budget microbatches. Its first pass used 500 warmup updates, peak learning rate 0.0003 and cosine decay to 0.00003. The second pass preserves optimizer state and starts at 0.00003, decaying toward 0.000003. Inputs and original targets are complete, without truncation. Earlier capped runs used different schedules; their settings are historical. Padding tokens do not contribute to loss. Model, optimizer, schedule and consumed-row state are saved regularly.

## Validation correction

Earlier validation excluded row positions but missed duplicated questions, so historical results were not fully independent. Before the current scratch run, a fresh holdout reserved 2,048 original question groups across shards 1-59. Every matching training row and tokenizer-shard question match is excluded. Two fixed 1,024-question subsets provide executed-answer diagnostics. The resulting stream contains 11,834,223 eligible original rows. Historical splits and their limitations are documented separately below.

# Quality experiments

## Historical runs

| Experiment | Evidence | Interpretation |
|---|---|---|
| Local MPS / short T4 checks | One local update; ten T4 updates; later T4 pilot reached 4,000 updates, loss 1.6469. | Pipeline checks. The longer pilot checkpoint was not recovered. |
| Sequential shards, 20,000 updates | Final held-out token loss 1.1213; best logged 0.9207 at step 12,500. | Best weights were not retained; last weights only. |
| Rotate after 1,000 mini-batches, 20,000 updates | Final loss 0.9815; saved best 0.8636 at step 18,500. | Selected historical checkpoint. Held-out exact code match 2/256; public score 18.85%. |

The historical comparison supports broader data coverage by token loss, but the validation-copy issue limits that conclusion. Token loss scores the next token with the reference prefix supplied; it does not show whether a generated program computes the right answer.

## Audit before the new pilots

All 1,024 held-out programs round-trip through the tokenizer exactly with cleanup disabled. Only 2 held-out questions and 3 of 1,919 test questions exceed the input limit; 5 held-out programs exceed the target limit. Truncation and tokenizer corruption do not explain most failures in this audit. Samples of 1,000 rows from shards 0, 15, 30, 45 and 55 have mean code lengths 65.96, 62.56, 93.17, 61.84 and 118.95 tokens. This suggests that sequential training encounters different styles for extended periods.

## Matched sampling pilot

Hypothesis: mixing original examples across shards improves held-out numeric agreement. Two fresh runs use identical initialization, model, seed, BF16 precision, batch 64, accumulation 1, grouping window 32 and 6,000 updates. Both exclude validation question texts. The control rotates after 1,000 batches. The treatment mixes up to 4,096 randomly selected original rows from every shard and resamples each epoch. Sampling is the experimental variable. Comparisons with older runs also change batch shape and validation exclusion.

| Run | Best token loss | Held-out numeric agreement |
|---|---|---|
| control | 1.6012 | 14/245 |
| mixed | 1.4793 | 16/245 |
| mixed_lr001 | 1.1409 | 27/245 |

Numeric evaluation compares solve() outputs of generated and supplied held-out programs, using limited separate processes and restricted imports. Only finite numeric reference results are usable. The old saved model agrees on 50/245 usable references (20.4%). Reference errors and runner restrictions can affect this measure. Numeric outputs are never used as training targets.

# Completed Drive-backed run

## All ten checkpoints retained

The mixed sampler completed 20,000 updates with batch 64, BF16 training and a peak learning rate of 0.001. Source commit: 1a23e45b1138c3dba7c40befeae6d3337153df98. Training plus periodic validation, generation and Drive saves took about 12 minutes on an NVIDIA RTX PRO 6000 Blackwell Server Edition. This is not a T4 timing measurement.

| Step | Validation token loss | Public correct / 960 | Public score |
|---|---|---|---|
| 2,000 | 1.8064 | 31 | 3.23% |
| 4,000 | 1.2130 | 49 | 5.10% |
| 6,000 | 1.0967 | 93 | 9.69% |
| 8,000 | 1.0020 | 103 | 10.73% |
| 10,000 | 0.9396 | 141 | 14.69% |
| 12,000 | 0.8848 | 133 | 13.85% |
| 14,000 | 0.8540 | 163 | 16.98% |
| 16,000 | 0.8219 | 182 | 18.96% |
| 18,000 | 0.8076 | 192 | 20.00% |
| 20,000 | 0.7952 | 195 | 20.31% |

Every checkpoint was written to Drive with optimizer state and a completion manifest before predictions were generated. Each checkpoint has 1,919 exact model outputs, matching CSV and JSONL files, validation metrics and provenance hashes. The files were exported from Drive, checked locally and then submitted to the bot. Colab was disconnected before the bot evaluations finished.

Drive location: MyDrive/HSE_NLP_HW2/runs/mixed-lr001-20k-drive-v1. The repository keeps exported prediction evidence and individual submission receipts. Weights stay in Drive. The final validation loss is lower than in the short pilots, but scores across checkpoints show that lower token loss does not guarantee higher program accuracy.

# Longer training and GPU use

## Eight more checkpoints, all scored

The continuation added 40,000 updates and 5,120,000 example presentations in 1,957.2 seconds (32.6 minutes), including validation, generation and Drive saves. It resumed our own step-20,000 model. Source commit: a411a64f8afe61e3a2b337af338c0ea6c19c9c15. No new target programs or external model weights were introduced.

| Step | Validation token loss | Public correct / 960 | Public score |
|---|---|---|---|
| 25,000 | 0.7647 | 194 | 20.21% |
| 30,000 | 0.7120 | 227 | 23.65% |
| 35,000 | 0.6817 | 231 | 24.06% |
| 40,000 | 0.6451 | 273 | 28.44% |
| 45,000 | 0.6058 | 314 | 32.71% |
| 50,000 | 0.5857 | 343 | 35.73% |
| 55,000 | 0.5701 | 353 | 36.77% |
| 60,000 | 0.5652 | 362 | 37.71% |

Batch size was chosen from two matched 16,384-example probes with reversed order. Median throughput was 2,100 examples/s at batch 64; 2,279 at 128; 2,114 at 256; and 1,950 at 512. Batch 128 was 8.5% faster than 64, with peak allocated memory 7.75 GiB. Larger batches raised utilization but reduced useful throughput.

Across the continuation, 1,972 one-second GPU samples averaged 60.38% utilization, with median 78%, 16.23% zero samples and peak used memory 10,397 MiB. These include checkpoint saving and generation. CPU preparation occupied about 41% of the batch-128 probe, suggesting a remaining bottleneck. The GPU was released and Colab showed no active sessions at 19:07 +03:00.

Drive location: MyDrive/HSE_NLP_HW2/runs/continue-20k-60k-b128-v1. Every checkpoint has exact 1,919-row outputs in official test order, provenance hashes and a bot receipt. The step-60,000 Blackwell generation took 9.07 seconds; its T4 timing remains unverified.

# Faster input preparation

## Matched CPU and GPU checks

CPU preparation compared the original batch path, reuse of window tokenization, and a bounded two-batch preparation queue. All three produced identical input IDs, masks and labels for 32,768 examples and 2,419,826 target tokens. Preparation took 18.72, 6.38 and 5.98 seconds respectively. Eighteen local checks covered truncation, duplicates, partial windows, literal special tokens and producer failures. Synthetic fixtures stayed confined to tests.

| Input path | Median examples / second |
|---|---|
| legacy | 2,931.17 |
| encoded | 3,483.24 |
| encoded-prefetch | 3,692.80 |

Two GPU rounds used reversed variant order, the same checkpoint, seed, examples, precision and batch size. The selected two-batch queue improved median throughput by 25.98% over the original path, with peak allocated tensor memory 7.75 GiB. The tokenizer used by the producer is separate from the validation tokenizer. This is a matched speed result; utilization alone does not measure useful throughput.

## Longer continuation: 60k to 180k

The run resumes our own verified 60,000-step checkpoint and schedules 120,000 additional updates (15,360,000 example presentations). It keeps the mixed sampler, batch 128 and BF16 training. Source commit: ca26869455692cb570b94cb4a633be7701221f9e. Weights and exact predictions are saved to Drive every 20,000 updates before export and bot evaluation.

| Step | Validation token loss | Public correct / 960 | Public score |
|---|---|---|---|
| 80,000 | 0.5250 | 414 | 43.13% |
| 100,000 | 0.4909 | 455 | 47.40% |
| 120,000 | 0.4712 | 472 | 49.17% |
| 140,000 | 0.4528 | 487 | 50.73% |
| 160,000 | 0.4419 | 516 | 53.75% |
| 180,000 | 0.4402 | 517 | 53.85% |

The saved completion record reports return code 0 and 6 checkpoints. All six were scored. The last 20,000 updates added one public correct answer (516 to 517). Training was stopped at its planned endpoint, GPU release was verified, and subsequent file work ran on CPU.

Each export is checked against its Drive manifest, source commit, checkpoint hash, official question-order hash and exact decoded selected token IDs. Bot receipts are written and read back on Drive. The current-generation measurements use Blackwell and do not establish T4 compliance.

# Generation and speed

## Measured optimization experiments

The course materials motivated length grouping, mixed precision, gradient accumulation, clipping and caching. The model already uses cached autoregressive generation. Training disables that cache. Loss accumulation stays on the GPU until logging. These choices were measured rather than assumed to make the full workload faster.

| Blackwell inference, same weights | Median seconds | Changed outputs |
|---|---|---|
| Before, FP32 | 11.350 | 0 |
| After, FP32 | 11.240 | 0 |
| After, FP16 | 13.027 | 54 |
| After, BF16 | 12.249 | 313 |

Two repetitions used reversed variant order on the second pass. The approximately 1% FP32 difference has overlapping timing ranges; it is not convincing evidence of a speedup. Lower inference precision was slower and changed programs. FP32 remains the default. An 80-update continuation benchmark also found no clear speed or validation-loss improvement; it does not establish full retraining quality.

## T4 requirement

The historical 39.09-second T4 measurement belongs to an older run. Blackwell timings cannot establish the T4 requirement for the current candidate. A bounded final-checkpoint T4 experiment is reported separately below; only its measured configurations can establish compliance.

## Exact model output and submission

Generation sorts questions by token length and restores their original positions. Historical runs followed greedy decoding with four-beam generation for syntactically invalid outputs. As of 16 October, strict greedy is selected: one candidate, no sampling, no syntax fallback, and no execution voting. No candidate is rewritten. Selected token IDs are saved, then decoded again to check equality with every output string. A sidecar records source commit, checkpoint hash, question-order hash, output hash, device and generation settings. JSONL conversion only wraps each exact string as a code field.

The historical submission included one handwritten identifier repair, which does not meet the user's stricter model-only condition. That repair has been removed from all future inference. The old score is retained as historical evidence, not a score for a newly verified model-only candidate.

The user confirmed that the required input is the official test.parquet question column in its original 1,919-row order. Generation preserves that order. The separate pass1_questions.jsonl is no longer a prerequisite.

# Agent use and reproducibility

## What the agent did

Codex read the assignment and course weeks, wrote data access, training, inference and diagnostic code, controlled Colab through MCP, inspected GPU state, compared measurements, prepared Telegram file packaging and wrote this report. It identified incomplete validation exclusion, stale T4 claims and handwritten postprocessing. Those findings changed the data exclusion rules, provenance records and speed checks.

Codex did not provide solution programs as training targets or replace model answers with its own solutions. No external model was used as a teacher. The submitted programs must come from the trained Transformer. Bot examples are not used to tune individual answers. The humanizer skill is used only to make report prose direct; it does not conceal agent involvement.

## What remains open

The early pilot weights were lost when temporary runtime storage was released before export succeeded; their measurements remain saved. The revised workflow passed CPU save and restore checks and retained all ten initial, eight first-continuation and six longer-continuation checkpoints, exact predictions and bot receipts in Drive. The training GPU was released. A child-process release request failed at the notebook disconnect step; the new inference notebook executes its final release from the notebook kernel and has a hard process timeout.

Longer training exceeded the 33% public baseline, but the first-place objective remains unmet. The best candidate and all intermediate checkpoints have actual bot scores. Changed decoding outputs require a new quality comparison; speed alone does not establish correctness. A public first-place result would not establish private first place. No private result is claimed.

## Reproduction files

Repository: OsVox/hse-nlp, branch 2026, homeworks/hw2. The directory contains train.py, data.py, numeric_eval.py, inference.py, tokenizer files, saved model weights, inference and training notebooks, experiment records and submission receipts. quality_experiments.md specifies the controlled pilot before its run. The raw logs and run configurations accompany the quality results.

Use a fresh output directory for each experiment. Run training with the recorded settings and keep latest plus best checkpoints. Run numeric evaluation on reserved training rows only. Use hw2_inference_colab.ipynb on a T4 to generate and package the test outputs. Verify every checkpoint in Drive before releasing compute; export predictions and record bot receipts from Drive.

## Sources

Course assignment; allowed dataset and schema; course week 2 (encoder-decoder attention and decoding), week 3 (training methods), week 4 (attention and caching). Local measurements: metrics.json, gpu_benchmark/*.json, quality_results.json when present, and submissions/2026-10-15-gpu-check.json.

# Alternatives to more updates

## Held-out decoding and checkpoint averaging

Both experiments used reserved training questions, with no public answers or bot failure examples used for selection. The numeric runner executes supplied references and exact model outputs with time and resource limits; 245 of 256 references were usable. Agreement is a diagnostic, not the course bot score.

| Experiment | Numeric agreement | Generation seconds, CPU | Decision |
|---|---|---|---|
| 120k greedy | 120/245 | 10.512 | Retain |
| 120k four beams | 125/245 | 57.190 | 5.44 times slower; not adopted |
| 140k unchanged | 122/245 | 10.160 | Retain |
| Average of 120k + 140k | 124/245 | 10.313 | Mixed evidence; not adopted |

Averaging used identical architecture, tokenizer and training-run settings. Parent weights were verified and averaged in CPU float64, then cast back and reopened. It adds no training targets or extra inference model. Validation token loss worsened from 0.452825 to 0.459383, and execution failures rose from 16 to 22. The two-answer diagnostic gain does not justify replacing the parent.

## Scaling-law estimate

The existing train-only probe contains 15,358 rows, sampled about equally across the 60 shards. With our tokenizer and length limits, mean question length is 46.91 tokens and mean code length is 73.83. Extrapolation gives approximately 555 million source tokens and 874 million target tokens per corpus. The last shard is smaller, so these are rough, non-deduplicated estimates. About 0.15% of sampled questions and 1.94% of sampled programs exceed the length limits.

Chinchilla suggests about 20 training tokens per parameter as a rough compute-efficient guide. Dividing target tokens by 20 gives 44 million parameters; counting source plus target gives 71 million. These are two accounting proxies, not a confidence interval or a fitted optimum for encoder-decoder code generation. Encoder and decoder capacity have different scaling behavior, and lower token loss need not imply higher task accuracy.

Proposed first larger model: width 512, feed-forward width 1536, six encoder layers and four decoder layers, totaling 42,481,152 parameters. Keeping four decoder layers limits repeated work during generation. At unchanged efficiency and time proportional to parameter count, one corpus-sized amount of data would take roughly 3.8-4.4 hours on the measured training GPU. This is a planning estimate; a short throughput/quality pilot and actual T4 timing must precede long training.

Hoffmann et al., Training Compute-Optimal Large Language Models; Ghorbani et al., Scaling Laws for Neural Machine Translation. Reproduction: scaling_estimate.py and experiments/scaling-analysis/.

# Final-checkpoint T4 inference

## Protocol

Prepare and verify model weights, tokenizer, official 1,919 questions and source on CPU. Save a hash-verified bundle to Drive, export and verify its ready record, then allocate one T4. Compare FP32 batch sizes 64, 128 and 256, twice with reversed order. Keep greedy generation, cached decoder state, length sorting, syntax-only beam retry and input/output limits 128/192 unchanged. Save every exact output, token provenance, hash and measurement to Drive. The two-model run omits optional profiling to keep GPU use bounded.

Generation time includes tokenization, sorting, generation, retries, decoding and local CSV writing. Full-call time additionally covers test reading, model hashing, provenance writing and verification. Model load and a small warm-up are measured separately. Drive archival is outside the generation timer. The experiment has a 600-second soft budget and a bounded child-process timeout; the notebook requests GPU release in finally.

| Model | Batch | Median, s | Slowest, s | Same outputs as 64 |
|---|---|---|---|---|
| 9.97M, 180k | 64 | 28.885 | 29.122 | True |
| 9.97M, 180k | 128 | 16.676 | 16.738 | True |
| 9.97M, 180k | 256 | 10.586 | 10.791 | True |
| 42.48M, 6k | 64 | 37.032 | 37.585 | True |
| 42.48M, 6k | 128 | 23.146 | 23.502 | True |
| 42.48M, 6k | 256 | 20.304 | 20.556 | True |

Device: Tesla T4; PyTorch 2.11.0+cu130; Transformers 4.57.6. Model loading took 0.289 seconds and warm-up took 0.954 seconds. Changed programs require separate quality evaluation; byte-identical outputs preserve the same predictions.

Batch 256 is selected for these immutable weights. Both models finish far below 180 seconds. Every run matches its already-scored JSONL exactly: 53.85% for the established model and 18.96% for the 6k larger pilot. Duplicate bot submissions were avoided. Later weights need a fresh timing and output check.

Full runs, per-call times, memory peaks, provenance and output hashes are retained in experiments/t4-comparison-v1 and its Drive archive. Reproduce with hw2_t4_comparison_colab.ipynb; the main inference notebook now selects batch 256 for the established checkpoint. The completed T4 runtime was manually released after its asynchronous release request left it connected.

# Short architecture and rate pilots

## Four fresh models, 2,000 updates each

The screen kept seed 42, batch 128, BF16 training, the same mixed data stream, tokenizer and 128/192 token limits. All runs used 200 warm-up updates and a short cosine schedule. Width, encoder depth and learning rate were varied in separate comparisons. One immutable checkpoint per trial was saved in Drive, exported with hashes and selected token IDs, verified locally, then scored once by the bot. Source: 1f1558515ce00d1be29f032bac843bcbfc5b6bf4.

| Parameters / layers | Peak LR | Val. loss | Held-out / 245 | Public / 960 | Train + export, s |
|---|---|---|---|---|---|
| 9.97M, 4/4 | .001 | 1.5262 | 18 | 66 (6.88%) | 106.67 |
| 35.66M, 4/4 | .001 | 1.9631 | 12 | 31 (3.23%) | 181.02 |
| 42.48M, 6/4 | .001 | 2.0145 | 9 | 27 (2.81%) | 193.07 |
| 42.48M, 6/4 | .0003 | 1.6312 | 23 | 86 (8.96%) | 190.10 |

Keep the 9.97M model at .001 and the 42.48M model at .0003. The small model has lower token loss and lower cost; the larger finalist has higher held-out execution agreement, confirmed by the bot. Lowering the rate within the 42M architecture improved all quality measures. The larger high-rate models do not justify continuation from this screen.

These are early-learning results from one seed, not final convergence estimates. Even the best pilot remains below the established 180k model at 53.85%. The bot reported public rank 10/15 on 16 October; earlier rank 9/15 was an earlier snapshot. No private score is available. The later T4 comparison covers the established model and the larger 6k finalist.

## Time and next comparison

The driver took 718.51 seconds after data restoration; its launcher record spans 758.79 seconds including restoration and imports. Across 716 GPU samples, mean activity was 72.90%, median 97%, and peak memory 18,627 MiB. Activity is not achieved compute throughput; startup, I/O and execution evaluation contribute low samples. Batch-size speed still needs a matched benchmark.

Continue both finalists from 2k to 6k with their optimizer moments, batch 128 and respective learning rates. Both receive the same new deterministic sample stream, a restarted 4k cosine schedule and 200 warm-up updates. Save and score the 6k checkpoint for each. Check actual T4 inference before long training. The four-way GPU was released manually after the automatic notebook request left it connected; file work resumed on CPU.

Evidence: experiments/short-pilots-v1, its selection.md and export.json, four submission receipts, finalist_pilots.py and hw2_finalist_pilots_colab.ipynb. Full weights, optimizer states and exact predictions remain in Google Drive.

# Finalists after 6,000 updates

## Matched continuation

Both models resumed their own 2k weights and Adam moments. Batch 128, BF16, original targets, the same new mixed sample stream, a restarted 4k cosine schedule and 200 warm-up updates were held fixed. Peak rates remained .001 for 9.97M and .0003 for 42.48M. Source: 10db3cda45ba777e04e4195f003a6b1db5254b24.

| Model | Val. loss | Held-out /245 | Public /960 | Added train + export, s |
|---|---|---|---|---|
| 9.97M | 0.9463 | 38 | 154 (16.04%) | 183.73 |
| 42.48M | 0.9782 | 41 | 182 (18.96%) | 353.36 |

The larger model leads execution accuracy in both evaluations; the smaller model has lower next-token loss and lower cost. The public difference is 28 answers (2.92 percentage points), while the held-out difference is only three answers. One seed and early checkpoints cannot establish an optimal model size or statistical significance. Both remain below the established 180k model at 53.85%.

## Measured cost

The added training loop took 160.2 seconds for 9.97M and 333.3 seconds for 42.48M. The wider timings in the table include startup, validation, checkpoint saves and all-test generation. At the larger model's measured rate, a corpus-sized amount of examples would take about 2.1 hours of training loop time at batch 128, before evaluation and archival. This improves on the older 3.8-4.4-hour parameter-linear estimate but remains a short-run extrapolation.

Across 583 GPU samples, median activity was 96%, mean activity 72.15%, and peak used memory 18,631 MiB. Activity includes I/O and numeric evaluation and does not measure achieved compute efficiency. A separate batch probe uses the actual encoded/prefetched training input path, equal examples, reversed rounds and timing-boundary synchronization.

## Checkpoint and evaluation retention

Both immutable 6k checkpoints, optimizer states, predictions and validation records were saved to Drive. The evidence archive was created there before download. Local checks verified hashes, all 1,919 official question positions and exact decoding of selected token IDs. Each checkpoint was submitted once to the course bot. Receipts are retained separately from the pre-submission archive. No answer text was edited.

The G4 runtime was manually released after its asynchronous release request left it connected. Standard CPU was used to stage the established 180k model and the larger 6k finalist for actual T4 inference. A faster training batch must also pass a short equal-example quality check before being adopted for full training.

Evidence: experiments/finalist-pilots-v1, its selection.md and export.json, the two 6k submission receipts, experiments/t4-comparison-v1/ready.json and benchmark_production_batches.py.

# Larger model: speed and quality

The 42.48M model used one prepared G4 session for a speed probe, short quality checks and a capped continuation. All eight probe passes used 32,768 original examples and 2,419,826 target tokens; the second round reversed the batch order. Probe weights were discarded.

| Batch | Median examples/s | Peak allocated GiB |
|---|---|---|
| 64 | 1794.6 | 8.18 |
| 128 | 1623.4 | 15.78 |
| 256 | 1433.7 | 30.97 |
| 512 | 1319.9 | 61.34 |

Batch 64 was 10.55% faster than 128. Both quality pilots resumed the same own 6k weights and optimizer state, with fixed added example counts, BF16 and peak rate .0003. The predefined gate required no fewer correct held-out executions and token loss within 1% of the control.

| Batch | Added examples | Token loss | Held-out /245 | Public /960 |
|---|---|---|---|---|
| 128 | 262,144 | 0.8854 | 45 | 183 |
| 64 | 262,144 | 0.8779 | 41 | 182 |

Batch 128 was retained. Lower token loss at batch 64 did not compensate for its lower execution accuracy. The chosen lineage continued to 2,560,000 total example presentations, equivalent to 20,000 updates at batch 128; this is not a full pass over the 11.84M-row corpus.

| Continuation step | Token loss | Public correct /960 |
|---|---|---|
| 10,000 | 0.8731 | 135 |
| 15,000 | 0.6995 | 266 |
| 20,000 | 0.6167 | 320 |

The 10k public regression followed a learning-rate restart; causality was not established. The final held-out result was 75/245. The grouped driver took 27.05 minutes including probes, preparation, evaluation and Drive saves. Each checkpoint retained exact token provenance and a separate bot receipt; no program was rewritten.

## Decoding opportunity

On the established 180k model, four-beam decoding improved held-out numeric agreement from 131/245 to 137/245 and reduced execution failures from 19 to 11. CPU generation took 10.61 versus 50.37 seconds. The subsequent full test export scored 544/960 (56.67%), versus 517/960 (53.85%) for greedy decoding. All 1,919 questions took 503.50 seconds on CPU at batch 64, FP32. T4 timing for this output remains unverified and is deferred. Every program matches its saved model token decode; no program was rewritten.

Source: 85532a78132c37ceb9cbf3f6cfff5c478d69258f. Evidence: experiments/grouped-42m-20k-v1, experiments/decoding-180k and experiments/beam4-quality-180k-v1 (source d1aec6e). Results were downloaded with the authenticated local Drive CLI, avoiding Colab allocation for export.

# Full model and decoding checks

The 42.48M model resumes its own 20k weights and Adam moments. The original target was 100k updates; the user requested a shift away from scaling, so the run was intentionally stopped after saving 70k. The GPU session was released. Settings were batch 128, BF16, peak learning rate .00015, 500 warmup updates, cosine decay and the encoded/prefetched input pipeline. Halving the prior peak is a restart hypothesis, not a fitted optimum. The first 30k and 40k bot scores are review points.

Before allocating G4, CPU checks verified all dataset hashes, loaded the model and optimizer, checked finite optimizer moments and saved a verified Drive bundle. At startup the GPU showed 98% activity and 18,951 MiB used out of 97,887 MiB. This is an observed snapshot, not a utilization guarantee. Each 10k checkpoint saves weights, optimizer state, exact predictions and a separate bot receipt.

| Step | Token loss | Public correct /960 | Public % |
|---|---|---|---|
| 20,000 (parent) | 0.6167 | 320 | 33.33 |
| 30,000 | 0.5475 | 409 | 42.60 |
| 40,000 | 0.4904 | 471 | 49.06 |
| 50,000 | 0.4516 | 519 | 54.06 |

At 3.84M example presentations, the larger model scored 409/960, versus 273/960 for the smaller model at step 40k. Different learning-rate schedules and data streams prevent a causal size-only comparison. These are scored checkpoints, not a completed target run. These historical exports use greedy decoding with syntax-only beam retries. No test labels are used for training. Source: 69829cbe; evidence: experiments/full-42m-100k-v1.

## Execution agreement without new training

Using the fixed 9.97M, 180k model, generate four deterministic beam candidates and execute their solve() functions in the restricted diagnostic runner. Select an unchanged program from a strict majority of all four numeric answers; otherwise retain the first executable candidate, breaking ties by model rank. References enter only after selection, for evaluation. No program is rewritten or added to training.

| Selection | Rows 512:768 (247 usable) | Rows 768:1024 (246 usable) |
|---|---|---|
| Top beam | 141 | 136 |
| First executable | 142 | 136 |
| Strict majority | 142 | 138 |

Strict majority passed the predefined positive-gain gate on both subsets: three extra correct answers across 493 usable references. This small exploratory gain is not statistical significance. First-executable selection failed the confirmation gate. Candidate and selected token decodes were checked after Drive export. The historical validation-copy caveat still applies to the 180k lineage.

Both tests ran on four CPU cores of the 48-core training host with CUDA hidden; GPU training continued. The full-test export scored 548/960 (57.08%), four more than beam-only decoding; CPU generation and selection took 279.43 seconds. T4 timing is unverified. Inspiration: Wang et al., Self-Consistency; our deterministic beam experiment differs from their sampling method. Sources: f047d7d (pilot), 124e46d (test export). Evidence: experiments/candidate-vote-180k-v1 and candidate-vote-test-180k-v1.

# Selected policy: strict greedy decoding

The user selected greedy decoding on 16 October. All new evaluated exports use num_beams=1 and do_sample=False, with syntax retries and execution voting disabled. Historical results are preserved with their original settings. Training stopped after its durable 70k checkpoint; separate CPU export processes apply strict greedy to saved weights.

| Step | Token loss | Public correct /960 | Public % |
|---|---|---|---|
| 50,000 | 0.4516 | 519 | 54.06 |
| 60,000 | 0.4255 | 540 | 56.25 |
| 70,000 | 0.4135 | 554 | 57.71 |

The 50k strict greedy output scored 519/960 (54.06%), matching the public score of its earlier retry-enabled output. Two of the 1,919 output strings differed; the comparison also changed device and batch size, so this is not a controlled attribution of the difference to retries alone. CPU generation took 99.67 seconds at batch 64. This is not T4 timing.

The default-decoder regression test supplies malformed Python and verifies that it is preserved with no extra generation calls. Full exports verify every program against saved token IDs, the official question-order hash, checkpoint hash and source commit. Each output is first saved to Drive, exported with the authenticated CLI, checked again and submitted once. Receipts are read back from Drive byte-for-byte.

Source: 8dcb74d2164a30c7b3169dcf7adf76114139761f. Evidence: experiments/decoding-policy.json, experiments/strict-greedy-42m-v1 and subsequent strict-greedy-42m checkpoint directories. The main inference notebook follows this policy; the earlier voting notebook is retained only for historical reproduction. First place remains unmet.

# Checkpoint averaging test

Arithmetic averaging combines our own 60k/70k or 50k/60k/70k weights into one model of unchanged size. It adds zero training updates and still generates one greedy candidate. All runs use CPU, FP32, batch 64 and two threads. The original Transformer paper also used checkpoint averaging, although its decoding settings differed.

| Model | Pilot correct /247 | Execution failures |
|---|---|---|
| baseline | 142 | 17 |
| avg60_70 | 152 | 15 |
| avg50_60_70 | 141 | 14 |

| Model | Confirmation correct /246 | Execution failures |
|---|---|---|
| baseline | 135 | 10 |
| avg60_70 | 136 | 13 |

Predefined gate: strictly more numeric agreements and no more execution failures than 70k on both held-out samples. Selected candidate: avg60_70; confirmed: False. These are small exploratory samples, not a significance test or public pass@1. Public test answers never choose the average.

Both candidates and their parent hashes are retained on Drive. Original held-out training references are used only for evaluation, with no generated training targets or handwritten repairs. Source: 861ae9795086a67af530f8eb2fc873423f466fbd. Evidence: experiments/checkpoint-average-70k-v1. Reference: Vaswani et al., Attention Is All You Need, section 6.1.

Bot result for baseline: 554/960 (57.71%). Exact file: strict-greedy-42m-070000.jsonl. T4 timing remains unverified.

# Short matched loss-weight test

Two independent 2,000-update continuations start from the same own 70k weights and optimizer state. Both use batch 128, seed 42, identical sampled data, BF16, peak learning rate .00003, 100 warmup steps and the same cosine schedule. Control uses the original loss. Numeric2 gives twice the weight to 305 of 8,192 tokenizer pieces matching numbers or arithmetic operators. Original programs are unchanged; matching pieces in comments and strings also receive weight. This is a heuristic, not a syntax-aware objective.

| Arm | Pilot agreements | Confirmation agreements | Public correct /960 |
|---|---|---|---|
| control | 141/245 | 284/493 | 562 |
| numeric2 | 138/245 | 289/493 | 573 |

The predefined held-out gate requires more numeric agreements and no extra execution failures on both subsets. Confirmed: False. This is a short exploratory comparison on previously used validation questions; it is not statistical significance. Training losses across arms are not directly comparable. Uniform validation loss and executed-answer diagnostics remain comparable.

Five focused tests checked normalized weighting, ignored padding, piece selection and continuation scheduling. CPU checks verified all dataset files, model weights and optimizer moments before G4 allocation. Both checkpoints and exact greedy outputs were saved to Drive before local export, token verification and bot submission. The control scored 562/960 (58.54%) and numeric2 scored 573/960 (59.69%), an 11-answer public gain at equal training budget. Numeric2 was retained at that stage for reproducing the highest public score; it did not pass the held-out gate for extending training. Total experiment time was 453 seconds, median GPU activity 97%. Colab showed no active sessions after completion. No longer training run started. Source: 8755a8f4867683d940e9b34fa48365e3cf4f8c21; evidence: experiments/numeric-loss-2k-v1. T4 timing remains deferred.

# Broad short experiment screen

Eight matched continuations start from the same 42.48M model at step 70,000, including its optimizer state. Each gets only 1,000 updates at batch 128. Seven candidates change one factor relative to control: numeric token weighting, label smoothing, equal question weight, dropout 0 or .05, doubled peak learning rate, or excluding targets longer than 192 tokens. Targets are never rewritten. Filtering refills complete batches but necessarily changes which examples are seen.

| Arm | Held-out /738 | Failures | Public /960 |
|---|---|---|---|
| control | 416 | 40 | 562 |
| numeric2 | 417 | 42 | 562 |
| smooth005 | 425 | 44 | 555 |
| equal_question | 416 | 38 | 562 |
| dropout0 | 414 | 38 | 564 |
| dropout005 | 413 | 40 | 559 |
| lr6e5 | 419 | 40 | 559 |
| complete_targets | 423 | 45 | 555 |

All outputs use strict greedy decoding. The shortlist requires positive total held-out gain, no negative gain on either validation subset, and no increase in total execution failures. At most three candidates proceed to a second-seed test against control. Combinations need their own short comparison before a long run; individual gains are not assumed to add together.

Shortlist: none. These previously reused validation questions provide exploratory evidence, not an untouched estimate of generalization. Public bot scores are recorded separately; individual public failure examples are not used to design targets or repairs.

Seventeen focused checks passed before CPU verification and GPU allocation. Checkpoints, exact greedy predictions, source identity and GPU telemetry are saved to Drive before export. Source: 5de771fc79b3b1e84926aedd8292560e56d9fecd. Evidence: experiments/broad-screen-1k-v1. Label smoothing and dropout motivation: Vaswani et al., section 5.4. Their usefulness here is determined by the measured trials.

Driver duration: 16.64 minutes, including data restore, training, evaluation and saves. Colab allocation and Drive sign-in occur before this timer. No long run is launched automatically.

# Training distribution audit

Before another GPU experiment, scan all 60 original training shards on CPU and exclude every exact copy of the 1,024 held-out question texts. Count exact questions using BLAKE2b-128 hashes and draw a fixed sample of 2,048 original pairs per shard for tokenizer statistics. No training targets are created or rewritten.

| Measurement | Observed |
|---|---|
| Eligible original rows | 11,836,444 |
| Distinct question hashes | 10,920,597 |
| Excess duplicate rows | 7.74% |
| Rows belonging to repeated questions | 13.48% |
| Sample size | 122,880 |
| Questions exceeding 128 tokens | 0.173% |
| Programs exceeding 192 tokens | 1.946% |
| Median / mean program tokens | 65 / 74.48 |

Exact duplication and truncation affect a minority of rows. This does not prove filtering cannot help, but it does not establish a large data-cleaning opportunity. The audit is descriptive: it does not assess semantic correctness, normalized paraphrases, or the effect of changing the training distribution.

A separate code inspection found that grouped batches always run in ascending input-length order within each window. An optional deterministic shuffle now changes only batch order, preserving every batch tensor and its padding cost. Thirteen focused checks passed, including unchanged default behavior and exact shuffled batch membership. This option is prepared for a short two-seed comparison; no score gain or completed training run is claimed.

Audit computation: 59.14 seconds; CPU process, input restore and setup: 85.97 seconds. No GPU was allocated. Colab confirmed no active sessions after automatic release. The contemporaneous public leaderboard remains 8/15: 59.69% versus 78.33% for first place. Evidence: experiments/train-diversity-audit-v1; audit source: 6add585. The first-place objective remains unmet.

# Grouped batch order: two-seed test

Four continuations start from identical own 70k weights and Adam moments. Within each seed, the control processes length-grouped batches in ascending length order; the treatment shuffles these same completed batches within each 4,096-example window. Both retain identical batch membership, padding cost and original targets. Seeds 42 and 1042 change the data/dropout stream. Each arm gets 1,024 updates, batch 128, or 131,072 example presentations; the budget ends on a complete grouping window.

| Seed / order | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control_s42 | 137 | 281 | 41 | 561 |
| shuffled_s42 | 136 | 285 | 45 | 550 |
| control_s1042 | 138 | 284 | 49 | 560 |
| shuffled_s1042 | 135 | 286 | 48 | 549 |

The predefined gate requires positive total held-out gain in each seed, no negative gain in either subset for either seed, and no increase in execution failures pooled across seeds. Confirmed: False. Reused questions across seeds are not independent test observations, and all validation remains exploratory.

The shuffled order gained 3 held-out correct answers in seed 42 but lost 1 in seed 1042, with 3 additional pooled execution failures. Public accuracy fell by 11 correct answers in each pair. No long extension is justified. The retained best remains 573/960 (59.69%), rank 8/15.

All arms use unchanged architecture, original token loss, peak learning rate .00003, 50 warmup updates and strict greedy decoding. Fifteen focused tests passed, including exact preservation of shuffled batch tensors, unchanged default behavior, checkpoint boundaries and paired selection. Every checkpoint and test output is saved to Drive before export and verification. Source: 862fa6f491c6c1bd4c7977d688aa37deaf2c54db; evidence: experiments/batch-order-1k-v1.

Driver duration: 8.60 minutes, excluding allocation and authorization before the driver. Median sampled GPU utilization was 97%; mean was 63.79%, including restoration, CPU scoring and saving. Colab confirmed no active sessions after automatic release. No long extension was started.

# Individual digits: short adaptation test

The treatment adds individual digit boundaries before BPE merges while retaining the original learned vocabulary, token IDs, model architecture and target text. The control and treatment continue from identical own 70k weights and Adam moments for 1,024 updates each, with batch 128, peak learning rate .00003 and 50 warmup updates. Both use strict greedy generation. The same original example pool is consumed at the complete-window boundary; longer sequences change grouping and padding.

CPU audit: 122,875 original pairs sampled across all 60 shards, excluding held-out question copies. Both tokenizers recover every original text exactly; save/reload and unchanged vocabulary IDs passed. Question mean length rises from 46.97 to 51.85 tokens; program mean from 74.52 to 80.94. At the unchanged 128/192 limits, question truncations rise from 218 to 446 and program truncations from 2,375 to 3,005.

| Tokenizer | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 137 | 281 | 41 | 561 |
| digits | 124 | 239 | 47 | 511 |

The predefined gate requires improvement on both held-out subsets and no increase in total execution failures. Shortlisted for a second-seed test: none. No long extension is automatic. This tests adaptation of an existing BPE model, not a model trained from scratch with digit boundaries. Repeated validation is exploratory.

The treatment lost 55 held-out agreements and added 6 execution failures. Public accuracy fell by 50/960 (5.21 percentage points). Reject this adaptation; the best remains 573/960 (59.69%), rank 8/15. The original control reproduces previous correct counts but 11 of 1,919 program strings differ, so bitwise repeatability is not established.

Motivation: Singh and Strouse, Tokenization counts (2024), https://arxiv.org/abs/2402.14903. That paper concerns arithmetic tokenization and does not establish a benefit for this Python-generation task. No paper examples or additional targets enter training.

Eighteen focused checks passed before launch. Every checkpoint and its original test output was saved to Drive before export and exact verification. Training source: f9676760c37607102647c7d0b792810f81f1e8f1. Evidence: experiments/digit-screen-1k-v1.

GPU driver duration: 4.90 minutes, excluding allocation and authorization. CPU preparation took 149.47 seconds, including the 65.65-second tokenizer audit. Colab confirmed no active sessions after automatic release; T4 timing remains unverified.

# Focal token loss: short matched screen

Three continuations start from the same own 70k model and Adam state. Each receives 1,024 updates, batch 128, peak learning rate .00003, 50 warmup updates and unchanged original data, vocabulary and programs. The control uses cross-entropy. Focal1 and focal2 multiply each original target-token loss by (1-p) raised to gamma=1 or 2, where p is its predicted probability. The gradient includes this factor; alpha=1. The denominator remains the original valid-token weight. Greedy decoding is unchanged.

| Loss | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 137 | 281 | 41 | 561 |
| focal1 | 139 | 283 | 40 | 559 |
| focal2 | 138 | 278 | 42 | 561 |

The predefined gate requires improvement on both held-out subsets without more total execution failures. Shortlisted for second-seed confirmation: focal1. No automatic combination or long run. Focal modulation changes gradient magnitude as well as emphasis, so identical nominal learning rates do not imply identical effective steps. Reused validation remains exploratory.

Evaluation now saves the exact generated programs and original references before scoring. Separate CPU-only scorers run while later GPU work proceeds. Payload hashes and row order are verified before scoring; the scoring branch imports no torch and does not load model weights. Process failures propagate to the driver, and pending processes are cleaned up. Twenty focused checks passed, including focal values/gradients, padding, score preservation and failure handling.

The verified dataset and parent archive from the earlier CPU preflight are reused; new source and launch files are uploaded to Drive and read back before GPU allocation. Archive hashes are rechecked on restore. Checkpoints and original test outputs are saved to Drive before export and bot scoring. Source: 86c3bfb603eeb2f20c7e3b4cf7dd3f246b084d93; evidence: experiments/focal-screen-1k-v1.

Motivation: Lin et al., Focal Loss for Dense Object Detection, ICCV 2017. Its object-detection results do not establish a benefit for this sequence-generation task. No external examples or additional training targets are used.

Focal1 gains four held-out agreements with one fewer execution failure, but its public score is two answers below control. Focal2 fails the held-out gate. No long run is justified; focal1 is only a candidate for short second-seed confirmation. The best remains 573/960 (59.69%), rank 8/15.

Driver duration: 6.35 minutes, excluding allocation and Drive authorization. Median GPU activity was 97%, mean 65.68%. Only 3 of 34 samples during CPU scoring show nonzero GPU activity, none at 90%: overlap was mostly with setup, and speedup is not established. Colab confirmed no active sessions. All three bot receipts are saved and verified in Drive. T4 timing remains unverified.

# Focal loss: second-seed confirmation

The gamma-one candidate alone is repeated against control with seed 1042. Both new arms start from the original own 70k parent and Adam state, not from the first-seed results. Each receives 1,024 updates with the same batch size, learning-rate schedule, data pipeline and strict greedy decoding. Seed 42 results are reused without retraining.

| Seed / loss | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 42 / control | 137 | 281 | 41 | 561 |
| 42 / focal1 | 139 | 283 | 40 | 559 |
| 1042 / control | 138 | 284 | 47 | 560 |
| 1042 / focal1 | 138 | 286 | 46 | 563 |

The predefined confirmation gate requires positive gains on both held-out subsets in each seed, with no increase in total execution failures in either seed. Confirmed: False. Public scores do not enter this gate. The same held-out questions are reused across seeds, so this is a robustness check rather than an independent validation set.

Twenty-one focused checks passed, including rejection of zero subset gains, extra execution failures and a failing first seed. The source archive includes the exact first-seed selection evidence and records its hash. Unchanged data and parent archives reuse the earlier CPU preflight; source and launcher are uploaded and read back before GPU allocation. Each new checkpoint and its predictions are saved to Drive before export and bot evaluation.

Source: bbf58a9376348573d28a72b798113fa0d6bab6a9. Evidence: experiments/focal-confirm-1k-v1. No additional training examples, target rewrites or handwritten output repairs are used. No long run starts automatically. T4 timing remains deferred.

The second seed gains two held-out agreements with one fewer failure, but the pilot subset ties control, so the predefined confirmation gate fails. Public gains are -2 and +3 across seeds. No long extension is justified. Best remains numeric2 at 573/960 (59.69%), rank 8/15. Next proposed check: a CPU audit of consistency among original duplicate-question programs, without creating targets.

The driver took 269.42 seconds (4.49 minutes), excluding allocation and authorization. GPU median activity was 97%, mean 64.09%, peak sampled memory 21,309 MiB. Both checkpoints and bot receipts are saved in Drive. Colab Manage Sessions confirmed no active sessions after automatic release.

# Python tokenizer: CPU comparison

At the user's suggestion, fit four new tokenizers on 61,440 original question/program pairs sampled across all 60 shards. Evaluate on 15,304 separate exact question texts; remove all held-out question copies from fitting and this audit. The existing 8,192-token byte BPE is retained as a reference. It was previously trained on shard 0, so its compression comparison is not an untouched generalization evaluation.

| Tokenizer | Question mean | Code mean | Code >192 | Code delta |
|---|---|---|---|---|
| original8192 | 46.86 | 74.11 | 278 | +0.00% |
| balanced8192 | 46.50 | 73.30 | 250 | -1.08% |
| code8192 | 46.98 | 73.09 | 250 | -1.36% |
| python8192 | 83.65 | 86.21 | 556 | +16.34% |
| code16384 | 45.44 | 71.76 | 219 | -3.17% |

Balanced8192 sees each original question and code once. Code8192 repeats the unchanged original code three times per question during tokenizer fitting. Python8192 adds explicit lexical-style boundaries for identifiers, numeric strings, operators, whitespace and indentation. Code16384 doubles the vocabulary with the same code emphasis. No pretrained vocabulary, new examples or rewritten training targets are used.

Every candidate is serialized, reloaded and checked for exact recovery of questions and code, including significant whitespace. Unknown tokens, number fragmentation and truncation counts are recorded. Shorter sequences do not prove better executed-answer accuracy. A different vocabulary cannot silently replace token meanings in the trained checkpoint; any deployment requires a matched training experiment.

Five focused local checks passed, including saved tokenizer round trips with tabs, CRLF and Unicode, disjoint question sampling, unchanged original Parquet files and separation of numeric conflict from unsupported execution. Motivation: Hugging Face CodeParrot methodology, https://github.com/huggingface/blog/blob/main/codeparrot.md. That work motivates a code-specific tokenizer; it does not prove a gain on this task.

Source: 52cccd19da35f70d023cf63f2aeb0d161524de18. Evidence: experiments/cpu-data-tokenizer-audit-v1/tokenizers. All work uses CPU; artifacts are saved to Drive before export.

The code-weighted 8k tokenizer shortens programs by 1.36% versus the current tokenizer, but only 0.28% versus the fresh balanced control. The 16k version shortens code by 3.17% and reduces over-limit programs from 278 to 219 of 15,304; at width512 its shared token table adds 4,194,304 parameters. The tested explicit Python boundaries lengthen programs by 16.34% and questions by 78.53%, so reject this particular boundary rule. All variants round-trip exactly with zero unknown tokens. Retain code8192 and code16384 as a matched vocabulary-size pair for a short future training test; compression does not establish an accuracy or wall-time gain. No tokenizer has replaced the best checkpoint.

# Original target consistency: CPU audit

Sample exact training-question groups using a fixed keyed hash, independently of public/test examples. Scan all original shards and exclude every held-out question copy. Among repeated questions with different original programs, test at most 512 groups and eight distinct programs per group. Execute only original programs with the existing restricted diagnostic runner; no values are converted into training targets.

| Measurement | Count |
|---|---|
| Eligible original rows | 11,836,444 |
| Sampled exact question groups | 10510 |
| Repeated question groups | 701 |
| Repeated groups with identical program | 198 |
| Repeated groups with different programs | 503 |
| Tested groups / programs | 503 / 1126 |
| Groups with program cap | 5 |
| all_executable_agree | 284 |
| numeric_conflict | 177 |
| insufficient_numeric | 38 |
| partial_agreement | 4 |

Numeric conflict means at least two executable original programs disagree within the existing 1e-6 tolerance. It does not identify which program is correct. Agreement also does not prove correctness. Unsupported or failed execution remains unresolved rather than an automatic reason to discard a target. Groups are sampled by question, not weighted by frequency; the result cannot be read as the fraction of all training rows that are wrong.

The audit saves question/program hashes, original shard/row positions, statuses and aggregate comparisons. It leaves original files unchanged and emits no training target file. Evidence: experiments/cpu-data-tokenizer-audit-v1/consistency.

Among 503 sampled repeated questions with distinct original programs, 177 (35.19%) have at least two conflicting numeric results; 284 fully agree, 38 have insufficient executable results, and 4 have partial agreement. Five groups hit the eight-program cap. This supports investigating a cheap original-data filtering experiment, but gives no authority to choose a correct answer or create targets. No filtering has been applied.

The complete CPU driver took 201.29 seconds (3.35 minutes), excluding allocation and Drive connection. No GPU was allocated. It exited successfully and Colab Manage Sessions confirmed no active sessions. Tokenizer artifacts, sampling provenance and audit results were saved to Drive before export and hash verification.

# Comments and empty lines: CPU preflight

At the user's suggestion, test removing non-executing text from existing training targets. The cleaner uses Python tokenization to distinguish comments from hash characters inside strings. Empty lines inside strings and explicit line continuations are preserved. Every accepted change must compile and have the same complete parsed Python structure, excluding source positions. Invalid or structurally changed programs fall back to the original.

| Measurement | Result |
|---|---|
| Sampled training programs | 4096 |
| Programs with comments | 503 |
| Changed / rejected | 2923 / 0 |
| Mean target tokens: original / cleaned | 65.80 / 62.77 |
| Target token change | -4.61% |
| Targets exceeding 192: original / cleaned | 23 / 10 |
| Executable pairs / identical numeric results | 249 / 249 |
| Numeric / status mismatches | 0 / 0 |
| Unsupported execution pairs | 7 |

The fixed sample comes from training shard 0 and excludes every exact copy of held-out question texts. This is a single-shard compression diagnostic, not evidence of improved model accuracy. Docstrings, pass statements, imports, assignments, variable names and calculations remain intact. Source or line-number introspection is outside the equivalence claim.

The training experiment replaces each original target once; it creates no additional examples, alternate solutions or numerical-answer targets. Generated submissions remain exact greedy model output, with no cleanup after generation. Nine focused cleanup tests cover quoted hash characters, multiline strings, docstrings, Unicode, line continuation, invalid syntax, repeatability and preservation of the input example stream.

# Representation: four short scratch trials

Compare original-tokenizer control, code-focused 8k vocabulary, code-focused 16k vocabulary, and comment/empty-line cleanup with the original tokenizer. Each receives 2,048 updates, batch 128, seed 42, learning rate 0.0003, and the same 262,134 original-example presentations (one batch has 118 rows). All models initialize at vocabulary 16,384; the 8k arms then shrink their tied embeddings. Their initial Transformer body hashes must match exactly. The dropout random stream is reset after shrinking.

| Arm | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 17 | 28 | 243 | 70 |
| code8192 | 20 | 36 | 190 | 86 |
| code16384 | 19 | 36 | 220 | 82 |
| comments | 17 | 34 | 217 | 65 |

The 8k models have 42,481,152 parameters; the 16k model has 46,675,456. Source stream and grouping-window budgets match, but changed lengths can change sorting inside a window. Token-level losses across vocabularies or target representations are not directly comparable. Evaluation uses original held-out programs and strict greedy decoding.

Predefined gate: improve both held-out subsets and add no total execution failures versus control. Candidates for a second-seed check: code8192, code16384. This early screen does not establish eventual performance after full training. Public results are recorded separately and do not enter the selection gate.

Source: 558c3fd580c18d8f7a1ca4a68a8126138fd3e062. Thirty-two checks passed before allocation. All checkpoints and predictions are saved to Drive before local export and bot submission. A hard runtime bound and automatic release are installed. No long run is started automatically.

Code8192 gains 11 held-out agreements and removes 53 execution failures; code16384 gains 10 and removes 23. Both improve each held-out subset and qualify for short second-seed checks. Comments gains six agreements but ties one subset, missing the strict gate; its public score also falls from 70 to 65. Do not combine cleanup or extend it into a long run on this evidence. Prefer the cheaper 8k tokenizer for confirmation first. Public scores are 70, 86, 82 and 65; all are early scratch models, not replacements for the retained 573/960 best checkpoint (rank 8/15). No full training starts.

The experiment driver took 858.86 seconds (14.31 minutes), excluding allocation, Drive reconnect and dependency setup. Colab showed an allocated GPU period of roughly 13:26–13:43 (minute precision). GPU median activity was 97%, mean 73.38%, peak sampled memory 22,739 MiB. The runtime was released automatically; Manage Sessions showed no active sessions. All four checkpoints and receipts are stored in Drive. T4 timing is unverified. At logged step 2,000, cleanup took 190.8 seconds versus 167.2 for control, so this implementation did not speed training. These training-loop times include CPU input preparation and exclude checkpoint export.

# Code-focused 8k: second-seed check

Repeat the cheaper code-focused tokenizer against the original 8k control. Both new models start from random weights with seed 1042, receive 2,048 updates and use the same data, architecture, batch size, learning-rate schedule and strict greedy decoding. The tokenizer files are unchanged from the CPU fit and verified byte-for-byte. The initial Transformer bodies are identical within each seed; no target cleanup is applied.

| Seed / tokenizer | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 42 / control | 17 | 28 | 243 | 70 |
| 42 / code8192 | 20 | 36 | 190 | 86 |
| 1042 / control | 18 | 39 | 197 | 78 |
| 1042 / code8192 | 23 | 37 | 205 | 73 |

The confirmation gate requires strictly positive gains on both held-out subsets in each seed, with no increase in total execution failures in either seed. Confirmed: False. Both seeds reuse the same held-out questions, so this checks seed robustness rather than independent validation or statistical significance. Public feedback is recorded separately and never converted into training targets.

The first-seed evidence is copied into the source archive and hash-checked before the new run. Eighteen focused checks passed before allocation. Existing verified dataset, tokenizer and model-shape preflight evidence is reused without allocating a CPU Colab runtime. Source, launcher and manifest are saved to Drive and downloaded back before GPU allocation. Each completed checkpoint and exact model output is saved to Drive before bot evaluation.

Source: e463ac8cb58f6bb52608f6d3e067e3c1739020e5. Evidence: experiments/tokenizer-confirm-2k-v1. A 900-second driver bound and automatic runtime release are installed. A positive result permits considering a bounded extension, not an automatic full training run.

The code-focused 8k tokenizer does not pass the predefined second-seed gate. It gains five pilot agreements but loses two confirmation agreements, with eight more total execution failures. Public score is 73/960 versus control 78/960; seed 42 had 86 versus 70. This small, mixed result does not justify a long run or combination with cleanup. Retain the 573/960 (59.69%) best checkpoint; latest bot rank is 8/15. No full training starts.

The two-arm driver took 431.28 seconds (7.19 minutes), excluding allocation, Drive and setup. GPU median activity was 97%, mean 73.99%, peak sampled memory 19,147 MiB. The launcher exited successfully and released the runtime automatically; Manage Sessions showed no active sessions. Both checkpoints, exact greedy outputs and bot receipts are stored in Drive. T4 timing is unverified.

# Original-data sampling: CPU preflight

Augmented question-program pairs are not used: the assignment prohibits additional training examples or targets. This experiment instead prepares inverse-frequency sampling of unchanged original rows. A question appearing f times is accepted with probability f to the power minus alpha. Alpha zero keeps the original sampler; 0.5 mildly reduces repetition; 1 applies full inverse-frequency weighting. All original program variants remain eligible. No answer is selected as correct and no program is rewritten.

The CPU index covers 11,836,444 eligible rows and 10,920,597 exact question strings across all 60 training shards. Every copy of a held-out question is excluded. One uint32 count is stored per original row; source and index checksums are verified before training. The existing mixed sampler draws equal candidates per shard, so this does not promise perfectly uniform question probabilities.

| Sampling power | Accepted rows | Unique questions | Expected repeat share |
|---|---|---|---|
| 0 | 245,750 | 244,692 | 13.44% |
| 0.5 | 233,833 | 233,649 | 9.03% |
| 1 | 226,787 | 226,719 | 6.22% |

The table measures one fixed candidate pool, not model accuracy or equal-budget training. Weighted pools are smaller; a future fixed-update trial draws further original rows to fill the same number of training batches. Candidate selection uses no public test information. Prior numeric disagreements do not identify the correct reference, so they are not used to delete or replace targets.

Five focused tests cover exact cross-shard counts, held-out exclusion, checksum rejection, deterministic probabilities, original target preservation and unchanged zero-power behavior. Source: 1abd5e36874b45a5f9c28e8c4d06b3a0da617074. No GPU is allocated for this audit.

| Power | Original rows | Unique questions | CPU loading seconds |
|---|---|---|---|
| 0 | 131,072 | 130,717 | 21.42 |
| 0.5 | 131,072 | 131,007 | 24.16 |
| 1 | 131,072 | 131,047 | 14.79 |

This timing covers data loading and sampling only, without tokenization or model training. It is not GPU throughput. Unique counts in a short stream need not increase: repeatedly occurring questions can fall outside the selected prefix by chance.

The index is ready for a bounded matched sampling trial, not a long run. Full inverse-frequency sampling reduces expected exposure to corpus-repeated questions from 13.44% to 6.22% in the audited candidate pool. At equal 131,072-row budget, within-stream repeated copies fall from 355 to 25; this adds only 330 distinct questions, so a large accuracy gain should not be assumed. Targets remain unchanged and no conflicting reference is chosen as truth. The next preregistered comparison uses the best 72k checkpoint, numeric-token weight 2 in every arm and 1,024 updates per arm after CPU parent/resume preflight. There is no new model or bot score from this audit.

The CPU driver completed in 141.71 seconds (2.36 minutes), excluding allocation, Drive mount and dependency setup; index construction took 48.89 seconds. Sampling 131,072 rows took 21.42, 24.16 and 14.79 seconds for powers 0, 0.5 and 1 in that order. These are single, ordered CPU data-loading measurements affected by caching, not evidence of training speedup. All 60 exported count arrays passed their checksums. The runtime released automatically and Manage Sessions showed no active sessions. No GPU was allocated.

# Inverse-frequency sampling: short training test

Three matched 1,024-update continuations from the verified 72k numeric-weight-2 checkpoint. The control preserves original mixed sampling. Inverse_sqrt and inverse accept an original row with probability 1/sqrt(f) and 1/f, where f is its exact question frequency across the training split. No input or target text is changed, and no reference is selected as correct. All arms retain numeric-token weight 2, the same optimizer state, batch size 128, learning rate 3e-5, 50-step warmup and strict greedy decoding.

| Sampling | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 139 | 286 | 40 | 575 |
| inverse_sqrt | 136 | 282 | 44 | 575 |
| inverse | 134 | 282 | 42 | 558 |

A candidate must improve both held-out subsets and add no total execution failures relative to control. Shortlist: none. A survivor needs a second-seed check before a longer run. Public scores are recorded separately and do not enter this selection rule. Repeated use of these validation questions and selection of the parent by public score make these results exploratory.

Eleven focused tests passed before allocation. CPU preflight verifies the parent weights, optimizer shapes and finite moments, resume step, a finite forward pass, the 60-shard frequency index and full batches at the actual continuation seed. Every model checkpoint and exact generated output is saved to Drive before export and bot evaluation. Training source: 0a3e2bfac3da70a5750efcf7233aa7506a6350e0. CPU data/model preflight used unchanged source 1d7ce0c3ec3ceae7d65e8d5153d308d5a5f69b8f. The first GPU attempt ended before training and left no checkpoint; Drive disconnected and its log was empty. Compute release was verified. The repaired launcher keeps its driver log locally and always requests compute release even if Drive export fails.

Reject both inverse-frequency treatments for longer training. Relative to control, inverse_sqrt loses 7 held-out agreements and adds 4 execution failures; inverse loses 9 and adds 2. Public results are 575/960 control, 575/960 inverse_sqrt and 558/960 inverse. Retain the simpler control for reproducing our new best 59.90%; the two-answer gain over its 72k parent is exploratory and does not establish a benefit from longer training. No long run or second seed is justified by this screen.

The successful three-arm driver took 367.16 seconds (6.12 minutes), with median GPU utilization 97%, mean 67.97% and peak memory 25,009 MiB. This excludes allocation/OAuth before the driver; the first failed attempt added approximately three minutes. CPU preflight took 122.83 seconds. Colab showed no active sessions after completion. T4 timing remains deferred.

# Comment cleanup after learning: short matched trial

The earlier 2,048-step scratch experiment cannot establish whether removing comments helps an already trained model. This experiment compares two 1,024-update continuations from the same verified 72k numeric-weight-2 parent. Control retains original targets. Comments removes comments and empty lines outside strings only when parsing and compilation succeed and the Python syntax tree is identical. Each original target is replaced once; no examples or alternate solutions are added. Strings, docstrings, variable names, imports and executable statements are preserved.

| Targets | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 139 | 286 | 38 | 575 |
| comments | 142 | 286 | 41 | 560 |

Both arms keep the original tokenizer, architecture, optimizer state, seed 42, batch 128, numeric-token weight 2, learning rate 3e-5, 50-step warmup and strict greedy decoding. Equal updates and complete batches preserve the example budget; target lengths change. Held-out references remain original. Shortlist: none. A candidate needs positive gains on both held-out subsets with no extra total execution failures, followed by another seed before any longer run. Public scores do not determine this gate.

Thirteen focused tests passed. Verified parent/data CPU checks and the prior cleanup audit were reused; no additional CPU Colab runtime was allocated. Source, launcher and readiness records were saved to Drive and downloaded back before GPU allocation. Every checkpoint and exact test prediction was exported from Drive and verified before bot evaluation. The updated finalizer saves local diagnostic logs and requests release even if Drive fails.

Do not extend comment cleanup. The trained-model cleanup arm scores 560/960 (58.33%) versus 575/960 (59.90%) for its matched unchanged-target control: 15 fewer public answers. It gains 3 agreements on the first held-out subset, ties the second and adds 3 total execution failures, so it fails the preregistered extension gate. Together with the earlier scratch pilot, these two settings provide no evidence for longer comment-cleanup training. This does not prove every possible cleanup schedule would fail. Keep the existing best control checkpoint at 59.90%; current public rank remains 9/16 at 15:08 (+03:00).

The two-arm driver took 286.86 seconds (4.78 minutes), excluding allocation, Drive authentication and launcher package setup. Median GPU activity was 95%, mean 56.51%, peak memory 25,901 MiB. At 1,000 added updates, the training log reports 86.0 seconds for control and 98.7 seconds for cleanup; on-the-fly cleanup did not improve training speed. Colab showed no active sessions after completion. T4 timing remains unverified and deferred.

# Sequence limits: CPU-only inference test

A CPU audit of the best checkpoint tokenizer found only 3 of 1,919 test questions longer than the 128-token input limit (maximum 166). The first 1,024 held-out questions include 2 above this limit, and 5 reference programs exceed the 192-token target limit. In the exact scored best export, 21 generated sequences lack an end-of-sequence token. One row overlaps, giving 23 affected rows.

Keep the same 42.48M model weights. Regenerate exactly those 23 rows on CPU with a 192-token input limit, a 384-token output limit, FP32, batch 8 and strict greedy decoding. Preserve the exact prior model-generated tokens on all other 1,896 rows. No code strings are edited, and no answer quality determines which result is retained. The full 1,919-row candidate is saved to Drive, exported and checked against generated token IDs before bot evaluation. This is a decoding test, with no training or GPU allocation.

Longer limits score 575/960 (59.90%), tying the retained best submission. They change 21 of 1,919 programs, but 5 of the 23 regenerated rows still lack EOS after the larger budget. Keep the existing inference notebook and checkpoint: this test shows no public score gain. Public rank is 9/16 at 15:25 (+03:00). No further extension is justified by this result.

The CPU driver took 44.31 seconds, including 19.72 seconds regenerating 23 rows; this excludes allocation, authentication and package setup. No GPU was allocated and no training was performed. A late monitoring request briefly reconnected a new CPU runtime after automatic release; that runtime was manually released, and Manage Sessions showed no active sessions. Incremental CPU timing is not a full 1,919-row inference measurement. T4 timing remains deferred and unverified.

Four focused checks passed. Model and tokenizer bytes are verified before CPU model loading. Each regenerated row is chosen solely from question length or missing EOS in the prior model output; no public reference answers enter the procedure. Source: 65f74d50bd9f6d68da0a555929783d0273a7d266. Evidence: experiments/sequence-length-audit-v1 and experiments/length-cap-cpu-v1.

# Decoder-input dropout experiment

A 15.20-second local CPU diagnostic used the existing best checkpoint and 256 reserved training examples. With correct previous target tokens, number/operator-piece prediction accuracy was 85.35%; pairing each target with another reserved question reduced it to 37.70%. This shows substantial question dependence. Removing all preceding-code embeddings reduced the same accuracy to 0.05%, but the interventions differ in severity: these numbers do not measure causal contribution shares or establish that dropout will help.

Test three matched continuations from the verified 72,000-step model: unchanged control, 10% and 20% whole-token decoder embedding dropout. Each adds 1,024 updates at batch 128, numeric-token loss weight 2 and peak learning rate 3e-5. Keep every supplied question and reference target unchanged. Only training-time decoder embeddings are randomly zeroed, with retained embeddings scaled to preserve their expectation. Preserve start/end/padding embeddings and use a separate random generator. Inference stays strict greedy with dropout disabled.

Reject both dropout rates for extension. Control scores 575/960 (59.90%), 10% dropout 546/960 (56.87%) and 20% dropout 542/960 (56.46%). Held-out agreement drops by 13 and 16 of 738 usable references, while execution failures rise by 18 and 21. Neither arm passes the predefined gate. Keep the existing best checkpoint and inference notebook at 59.90%, rank 9/16 as reported at 15:44 (+03:00). This short continuation result does not establish whether training with dropout from scratch would help.

The three-arm driver took 393.35 seconds (6.56 minutes), excluding allocation, Drive authentication and package setup. Median GPU utilization was 97%, mean 65.60%, peak memory 23,885 MiB. All checkpoints and exact greedy predictions were saved to Drive before local verification. The control exactly matches a previously scored JSONL, so its 575/960 receipt was reused without a duplicate submission. Colab Manage Sessions visibly showed no active sessions after automatic release. T4 timing remains deferred and unverified.

Five unittest checks preceded launch, covering dropout and continuation behavior. Two runtime-finalization checks were subsequently rerun explicitly with pytest (seven dropout/continuation/runtime checks in total). Parent optimizer/data preflight was reused with checksum verification. Inspiration: Zhang et al., Token Drop mechanism for Neural Machine Translation, COLING 2020, https://aclanthology.org/2020.coling-main.379/. Only whole-token embedding dropout is tested; no extra targets, teacher or auxiliary objective is adopted. Source: a4a79fb19b7a45ba1bcec3517309d11cfe1d8b02. Evidence: experiments/conditioning-audit-cpu-v1 and experiments/decoder-dropout-1k-v1.

# Learned source copying: CPU pilot

A 9.54-second CPU diagnostic on 256 reserved original examples found 207 number-piece prediction errors. The exact target token occurred in the question for 62 errors across 41 examples; an untrained average of all attention heads recovered 21 of those 62. These teacher-forced token diagnostics are not whole-program accuracy or a guaranteed achievable gain. Number pieces also include comments and strings.

Keep all 42,481,152 parent Transformer parameters frozen. Fit 545 parameters: a decoder-state copying gate and a learned mixture of 32 cross-attention heads. At every step, mix normal vocabulary probabilities with probabilities of copying existing question token IDs. Sum repeated source tokens, exclude padding/end tokens, and take the strict greedy maximum. No program repair, hand-written answer rule, generated target or pretrained component is used.

Use 4,096 original examples from supplied training shard zero, excluding all questions in the first 1,024 held-out rows. Fit for 512 small updates with the existing numeric/operator weight 2. Save and checksum-verify the custom CopyT5 checkpoint on Drive before generation, then reload the identical local weights; verify that every base parameter is unchanged. Compare copying enabled versus disabled on the same saved model and held-out subsets. Export all 1,919 test predictions in official order for bot evaluation.

Do not extend this frozen-model copying pilot. Control and learned copying both score 139/245 and 286/493 on the held-out subsets, with 40 total execution failures. The bot scores the new unique output 575/960 (59.90%), tying the simpler retained best. Public rank is 9/16 at 16:16 (+03:00). The positive-gain requirement fails; no second seed or long run is started. This tests a small frozen-feature adapter, not joint training of a Transformer and copy mechanism.

The CPU-only driver took 572.00 seconds (9.53 minutes), excluding allocation, Drive authorization and package setup. Fitting 545 parameters on cached frozen-model features took 1.93 seconds; all-1919 test generation took 179.83 seconds on CPU. Base weights remained identical. All checkpoints and predictions were saved and checksum-verified on Drive before local export and bot submission. The driver exited successfully; Colab showed no active sessions. No GPU was used, and T4 timing remains unverified.

Eleven focused pytest checks passed, including cached greedy generation, save/reload, copying normalization, gradient flow, disabled equivalence and runtime finalization. Real-checkpoint CPU preflight confirmed finite adapter gradients and no base gradients. Source: 0938a6265db552fd794e8e691cb30d646069ea60. Evidence: experiments/copyability-audit-cpu-v1 and experiments/copy-adapter-cpu-v1. Pointer-generator reference: See, Liu and Manning (2017), https://aclanthology.org/P17-1099/. Reused validation is exploratory; no long run is automatic.

# Full-model optimizer comparison from scratch

Four 42.48M Transformers start from identical random weights, with the same original 8k tokenizer, original example stream, seed 42, batch 128 and 2,048 updates. Every parameter trains. AdamW uses rates 0.0003 and 0.001. Muon uses rates 0.001 and 0.003 on hidden matrices; its embeddings and vectors use AdamW at 0.0003. Native PyTorch Muon uses five Newton-Schulz iterations, momentum 0.95 and match_rms_adamw scaling. Tied output/input embeddings and relative-position embeddings are excluded from Muon.

| Optimizer / rate | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| AdamW 0.0003 | 18 | 33 | 219 | 70 |
| AdamW 0.001 | 3 | 7 | 424 | 21 |
| Muon 0.001 | 22 | 39 | 188 | 96 |
| Muon 0.003 | 2 | 0 | 420 | 7 |

Choose the stronger AdamW rate by summed held-out agreements, breaking ties by fewer failures then lower rate. Muon must improve both subsets without extra failures. Each survivor requires another seed before any longer or combined run. This compares early fresh-model learning; it does not test switching optimizers on the mature model, and it does not establish optimal rates or model size. Public feedback is recorded separately.

Muon at rate 0.001 passes the first exploratory screen: held-out gains are +4/245 and +6/493 over the better AdamW rate (0.0003), with 31 fewer total execution failures. Muon at 0.003 and AdamW at 0.001 regress. Repeat the selected low-rate Muon recipe and AdamW control with seed 1042 before any extension; the result is not yet replicated. All weights trained from scratch. This is not a continuation of the 73k best checkpoint and does not replace it. The original best remains 575/960 (59.90%).

Each arm processed exactly 262,134 original examples, 12,278,660 source tokens and 19,293,376 target tokens in 2,048 updates. Training process times including setup/checkpoint generation were 177.76, 177.45, 201.15 and 201.47 seconds (AdamW low/high, Muon low/high). Low-rate Muon costs 13.16% more than control at matched examples. Total driver time including data restore, probe and held-out evaluation was 892.81 seconds (14.88 minutes); the conservative allocation-to-release wall-time bound is 1,171.93 seconds (19.53 minutes), not an exact billing measurement. GPU utilization over the whole driver averaged 75.36%, median 96%, peak memory 20,714 MiB. All four checkpoints and exact greedy outputs were saved to Drive, exported and verified. No active Colab session remains. T4 inference timing is unverified.

Twelve focused CPU checks passed, plus four discarded full-size one-step CPU preflights with identical initial loss and finite gradients for every model parameter. Colab CPU confirmed native Muon before GPU allocation. A discarded GPU timing probe estimated 0.104 seconds/step for AdamW and 0.115 for Muon on the same 128-example batch. Source: bb3c316097764701dc0eaa822b3eafc5decaf9b1. Native implementation: https://docs.pytorch.org/docs/2.9/generated/torch.optim.Muon.html. Inspiration: https://github.com/KellerJordan/modded-nanogpt; its language-model speedrun results do not imply gains on this task.

# Muon: second-seed replication

Repeat only the selected AdamW 0.0003 and Muon 0.001 recipes with shared seed 1042. The original screen used seed 42. Every parameter of each 42.48M encoder-decoder trains from scratch for 2,048 updates; the original 8k tokenizer, batch 128, original target loss, warmup and data pipeline stay fixed. Muon uses AdamW 0.0003 for embeddings and vectors. No frozen backbone, pretrained weights, generated targets or test feedback enters training.

| Recipe | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 20 | 38 | 185 | 72 |
| muon_low | 26 | 48 | 140 | 115 |

The candidate must improve both held-out subsets and add no total execution failures in both seeds. The prior selection file is checksum-verified before launch. These reused validation questions make the result exploratory; replication across seeds does not establish statistical significance, a compute-optimal model size or eventual mature-model accuracy. Equal examples also do not imply equal GPU cost.

The selected Muon recipe passes the two-seed screening rule. On seed 1042, it scores 26/245 and 48/493 versus AdamW 20/245 and 38/493, with 45 fewer total execution failures. Seed 42 gains were +4 and +6 with 31 fewer failures. The bot scores 115/960 (11.98%) versus 72/960 (7.50%) for the new pair; previous seed scores were 96 versus 70. This supports further bounded testing of Muon, not a guaranteed mature-model gain. Keep the existing 575/960 model. Latest bot rank is 10/16; first place observed at 78.33%. No long run has started.

Each arm processed 262,136 original examples, 12,284,754 source tokens and 19,280,571 target tokens in 2,048 updates. Training process times including setup and checkpoint generation were 177.79 seconds for AdamW and 200.35 for Muon (12.69% more). Driver time was 456.24 seconds (7.60 minutes); the conservative allocation-to-release upper bound was 568.07 seconds (9.47 minutes). Whole-driver GPU utilization averaged 73.83%, median 96%, peak memory 19,147 MiB. Both exact outputs were saved to Drive, exported and scored; remote model downloads were checked against saved hashes. Colab has no active sessions. T4 timing remains unverified.

Ten focused checks passed. Existing full-size CPU model/optimizer checks and native-Muon Colab compatibility evidence were reused because the implementations and rates are unchanged. Preparation files were uploaded to Drive and downloaded back before allocation. Each checkpoint and strict-greedy test output was saved to Drive before export and exact-token verification. Source: 310a4d89ec8fd62d58d59e7042134cd4bd7c23ac. Evidence: experiments/optimizer-confirm-2k-v1.

# Encoder/decoder capacity balance with Muon

Compare three fresh models using the replicated Muon 0.001 recipe, with AdamW 0.0003 on embeddings and vectors. All have model width 512 and eight attention heads. Encoder/decoder depths 6/4, 8/2 and 4/6 use feed-forward widths 1536, 1664 and 1408 respectively, keeping parameter counts within 0.32% of baseline. Each uses seed 42, the original 8k tokenizer and unmodified data, batch 128 and 2,048 updates. Different tensor shapes cannot share identical random weights; each initial model hash is recorded.

| Encoder / decoder | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 6 / 4 (42.48M) | 23 | 41 | 177 | 101 |
| 8 / 2 (42.35M) | 19 | 30 | 357 | 59 |
| 4 / 6 (42.61M) | 25 | 51 | 103 | 108 |

A candidate must improve both held-out subsets with no extra total execution failures versus the newly trained 6/4 Muon control, then repeat that gain on another seed. This changes depth balance and compensating feed-forward width together. Equal parameters and examples do not imply equal compute; architecture-specific learning rates were not tuned. Public scores are recorded separately and do not select candidates.

The 4-encoder/6-decoder candidate passed the first screen: +2 and +10 correct on the two held-out subsets, with 74 fewer execution failures. It requires seed-1042 confirmation before extension. The 8/2 candidate failed this equal-example screen (-4 and -11, 180 extra failures); this does not rule out a separate equal-time comparison. Public results were 101, 59 and 108 /960 for 6/4, 8/2 and 4/6. The mature 575/960 checkpoint remains the best.

The full three-arm driver took 701.81 seconds (11.70 minutes), including evaluation. Whole-driver GPU activity had median 95%, mean 74.36%, and peak reported memory 20,325 MiB. The runtime disconnected automatically after exporting results. These figures do not establish equal GPU cost across architectures.

CPU preflight found that Transformers 4.57.6 inferred the default T5 cache depth from encoder num_layers, raising IndexError for the 4/6 model. Deeper-decoder models now save native dynamic_full caching in generation_config. Cached and uncached greedy IDs agree in the focused regression test, including after save/reload. Full-size finite-gradient and cached-generation checks pass for all three shapes; 13 focused checks passed before allocation. No GPU was used for this fix. Source: f6fc269034b3e42ec3e3e8850ecab63d1e726d76. Evidence: experiments/depth-muon-2k-v1.

# Second-seed depth confirmation

Repeat the 6/4 control versus 4/6 encoder/decoder candidate using seed 1042. The original tokenizer, examples, unmodified target programs, batch 128, 2,048 updates and Muon/AdamW rates remain fixed. All model parameters are trained from scratch. The original seed-42 selection is checksum-verified before the confirmation run. This repeats the same nearly matched-parameter architecture comparison; shapes and compensating feed-forward width differ.

| Seed 1042 model | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 6 / 4 control | 24 | 47 | 145 | 102 |
| 4 / 6 candidate | 27 | 42 | 147 | 109 |

The 4-encoder/6-decoder candidate did not replicate its first-seed held-out improvement: seed 1042 gained 3/245 on the pilot subset, lost 5/493 on confirmation, and added 2 execution failures in total. Public bot scores were 102/960 for the 6/4 control and 109/960 for 4/6, but the predeclared held-out gate failed. Keep 6/4 for the next screen; do not start a long depth-change run. The mature 575/960 checkpoint remains the best. A repeated seed-1042 6/4 control varied from an earlier 115/960 to 102/960 despite matching recorded configuration and initial body hash; the cause is not isolated. Small gains remain uncertain.

The complete two-arm driver took 595.30 seconds (9.92 minutes), including dataset restoration, training and evaluation. Control process time was 202.43 seconds and deeper-decoder time 237.66 seconds (17.4% longer), with equal 262,136 examples per arm. Whole-driver GPU activity had median 95%, mean 66.19%, and peak memory 21,113 MiB. Both complete model files were downloaded from Drive and checksum-verified. Colab release was verified after automatic disconnection.

The candidate must improve both held-out subsets with no extra total execution failures in both seeds. The public bot score is logged separately. Reused validation makes this exploratory; replication does not establish eventual mature-model accuracy or statistical significance. Fifteen focused CPU checks passed before allocation; prior full-size finite-gradient and cache checks were reused because the model, optimizer, input and evaluation implementations are unchanged. Each checkpoint and exact greedy output was saved to Drive before export and bot submission. Source: 10520ebb061a9d7b4353bc1f1fa53664030b128b. Evidence: experiments/depth-confirm-2k-v1.

# GPU and batch-size efficiency

A disposable Muon training probe uses the same scratch 42.48M model, original tokenizer, BF16, and encoded input path with two prefetched batches. Each timed measurement processes the same 8,192 original shard-0 training examples, excluding held-out questions. Each size receives three warm-up updates, followed by ascending and descending measurement rounds. Useful token counts are equal; padding differs. Results are medians of two short measurements, not full-dataset training speed or evidence of better learning.

| GPU | Batch | Examples/s | Peak GiB | Units/hour |
|---|---|---|---|---|
| G4 | 128 | 1503 | 15.6 | 8.9 |
| G4 | 256 | 1453 | 30.8 | 8.9 |
| G4 | 512 | 1354 | 61.2 | 8.9 |
| A100 80GB | 128 | 848 | 15.6 | 6.77 |
| A100 80GB | 256 | 1028 | 30.8 | 6.77 |
| A100 80GB | 512 | 1111 | 61.2 | 6.77 |

Retain G4 with batch 128. Median throughput is 1,503 examples/s, versus A100's best tested 1,111 examples/s at batch 512: G4 is 35.3% faster. Batch 256 and 512 reduced G4 speed by 3.3% and 10.0%; batch 1024 ran out of memory on both GPUs. G4 has about 3% better measured examples per compute unit than A100's larger-batch winner, a small difference that should not be overinterpreted from a short probe. No training recipe or model weights changed.

Colab could not allocate the requested H100 and substituted G4. That duplicate probe was stopped and is excluded from this comparison. A100 allocation was verified as NVIDIA A100-SXM4-80GB. Quoted rates observed in this session were 8.9 units/hour for G4 and 6.77 for A100. Both runtimes were released; G4 is selected but disconnected.

All probe weights are discarded. The supplied code, prepared inputs, hashes, logs and measured results are preserved on Drive. Each runtime has a bounded worker and automatic release. A batch change must separately pass a short quality comparison before adoption. CPU checks verified equal useful token counts before allocation. Benchmark source: commit 9e1a489. Evidence: experiments/muon-batch-probe-v1 and the adjacent H100/A100 attempt folders.

# Muon transfer to the mature checkpoint

Four matched 1,024-update continuations start from the same own scratch-trained 73,024-step, 42.48M-parameter model (575/960 public). Compare AdamW with saved moments, reset AdamW, and reset Muon at matrix rates 0.0001 and 0.0003. AdamW on embeddings and vectors stays at 0.00003. Original tokenizer, examples, batch 128, numerical-token loss weight 2, 50-step warmup and cosine schedule remain matched. All parameters train. The reset AdamW arm separates the optimizer choice from discarding stored moments.

| Optimizer arm | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 136 | 283 | 40 | 568 |
| adamw_reset | 137 | 282 | 40 | 571 |
| muon_low | 133 | 286 | 45 | 569 |
| muon_high | 138 | 280 | 49 | 568 |

Neither Muon transfer rate passed the declared held-out gate. Relative to the stronger reset-AdamW reference, low-rate Muon changed correct counts by -4/+4 with 5 extra failures; high-rate Muon changed them by +1/-2 with 9 extra failures. Both also lost agreement on at least one subset against the parent. Public scores were 568, 571, 569 and 568 /960 for continued AdamW, reset AdamW, Muon 0.0001 and Muon 0.0003. Retain the parent 575/960 model and do not extend these transfers. This does not invalidate Muon gains from fresh initialization or establish that every transfer rate fails.

The four-arm driver took 535.41 seconds (8.92 minutes). Each arm processed 131,072 identical examples, 6,152,967 source tokens and 9,646,354 target tokens. Process times were 98.92, 94.22, 106.52 and 106.43 seconds. Whole-driver utilization was median 96%, mean 68.77%; peak memory was 21,443 MiB. Allocation-request to verified zero sessions was at most 845.21 seconds (14.09 minutes), including authorization, setup and cleanup. An extra cleanup cell raced automatic release and briefly reconnected an empty runtime; it was removed through Colab UI. After terminal worker status, use read-only Drive/UI checks, not executable diagnostic cells.

Selection compares Muon against the stronger AdamW arm on the two reused held-out subsets. Require positive gains in both and no additional total execution failures; also require no loss against the parent (139/245 and 286/493). Any survivor requires a second-seed comparison. Public scores are recorded separately. This small screen cannot establish optimal rates or statistical significance.

Eleven focused CPU tests pass, including optimizer-state reset, continuation schedules and selection gates. Full-size CPU forward/backward/update checks are finite for all four arms; the parent manifest is verified. Source f05c15878bb2be8bddf88f7385e48a580cbc84e7 and the parent archive were uploaded and downloaded back before GPU allocation. No pretrained weights, rewritten labels or test-reward optimization are used. Evidence: experiments/muon-transfer-1k-v1.

# Question lengths and context limits

CPU-only audit performed after the encoder/decoder confirmation was scored and archived. Tokenize original questions with the current original 8k tokenizer, without padding or truncation; counts include the final end token. All eligible shard-0 rows exclude exact held-out question texts. This is not a census of the full 60-shard corpus. The earlier 15,304-question sample across all 60 shards averaged 46.86 tokens.

| Questions | Count | Mean tokens | Median | 95th / 99th |
|---|---|---|---|---|
| Eligible shard 0 | 198952 | 43.66 | 40 | 78 / 105 |
| Official test | 1919 | 43.72 | 40 | 76 / 101 |

| Input cap | Train questions cut | Test questions cut |
|---|---|---|
| 64 | 23,166 (11.64%) | 230 (11.99%) |
| 96 | 3,280 (1.65%) | 27 (1.41%) |
| 128 | 517 (0.26%) | 3 (0.16%) |
| 192 | 20 (0.01%) | 0 (0.00%) |

Retain the current 128-token input limit pending a quality and speed comparison. A 64-token limit would truncate about 12% of test questions. Training already groups similar lengths in windows of 32 batches, and both training and inference pad only to the longest sequence in each batch; a 128-token cap does not make every input 128 tokens long. These counts do not measure GPU savings or establish the best cap.

The three test inputs above the current limit contain 137, 146 and 166 tokens including the end token. Input coverage is therefore a reason to test a larger limit as well as more efficient grouping. Neither the 128-token source cap nor the 192-token target cap changed. No model was trained, predictions generated or bot submission made for this audit. Original and mature-checkpoint tokenizer definitions agree after removing saved padding/truncation settings.

Evidence: context_length_audit.py and experiments/context-length-cpu-v1. The CPU measurement took 3.29 seconds; no Colab allocation was made. Full histograms, input and tokenizer checksums, and the earlier sample reference are retained.

# Short model-size comparison with Muon

Three fresh encoder-decoder Transformers use the retained 6/4 depth and 64-dimensional attention heads. Model/feed-forward widths and head counts are 320/960/5, 512/1536/8 and 704/2112/11. All parameters train for 2,048 updates with seed 42, batch 128, BF16, the original 8k tokenizer and unmodified course examples. Hidden matrices use Muon 0.001; embeddings and vectors use AdamW 0.0003. Validation token loss is recorded every 512 updates, while the single final checkpoint per arm is saved and evaluated with strict greedy decoding.

| Parameters | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 17.58M | 27 | 49 | 117 | 90 |
| 42.48M | 22 | 37 | 219 | 107 |
| 78.15M | 23 | 45 | 229 | 110 |

| Model | Process seconds | Val loss at 512 / 1024 / 1536 / 2048 |
|---|---|---|
| 17.58M | 137.7 | 2.485 / 1.879 / 1.601 / 1.498 |
| 42.48M | 202.9 | 2.167 / 1.692 / 1.468 / 1.389 |
| 78.15M | 283.8 | 2.045 / 1.659 / 1.471 / 1.395 |

Shortlist the 17.58M model for a second-seed comparison against 42.48M: held-out correct counts were 27/245 and 49/493 versus 22/245 and 37/493, with 117 versus 219 total execution failures. The 78.15M model improved to 23/245 and 45/493 but added 10 failures, so it failed the predeclared gate. Public scores disagree with the held-out ranking: 90/960 small, 107/960 control and 110/960 large. The small model is therefore a candidate for replication, not an established optimum. All runs used equal examples, not equal GPU time. No long run started; the mature 575/960 checkpoint remains best.

The three-arm driver took 729.70 seconds (12.16 minutes), including dataset restoration, training and held-out evaluation. Training process seconds: small: 137.70, control: 202.93, large: 283.77. Whole-driver GPU utilization mean 75.15%, median 95%; peak memory 25023 MiB. Each final checkpoint was saved to Drive, its model SHA verified by download, and all 1919 greedy outputs verified before bot submission. Allocation request to verified zero active sessions was at most 865.19 seconds (14.42 minutes), including setup and release verification.

The predeclared gate requires improvement on both held-out subsets and no extra total execution failures versus the fresh 42M control. Public scores are recorded separately. A survivor still needs another seed and a comparison at matched measured GPU time before longer training. Equal updates do not equal compute; rates were not tuned separately for each size. These short curves and reused validation questions cannot establish an optimal final model size.

Twelve focused CPU tests passed, plus finite-gradient and cached-versus-uncached greedy checks at all three actual model sizes. Files were uploaded to Drive and downloaded back before allocation. Every final checkpoint and generated output was saved to Drive before export, checksum verification and bot evaluation. Training source: 926d03d7834ef412e27c4a30bca173a978723c53. Evidence: experiments/size-muon-2k-v1.

# Training coverage: audit and correction

Access to all 60 source files did not establish a full pass over the training data. The old mixed sampler selected 4,096 rows from each shard per round and sampled again the next round. Rows could repeat before the rest had been visited. Resuming also selected a new sampling seed. Independent pilot models do not share their training experience.

| Checkpoint | Example presentations | Maximum distinct-row coverage |
|---|---|---|
| 78M at 2,048 updates | 262,134 | 2.21% |
| Best 42M at 73,024 updates | <= 9,347,072 | <= 78.97% |

These are upper bounds, not observed unique counts: repetitions reduce coverage. The denominator is 11,836,444 eligible original rows, excluding all copies of held-out questions. Some visited examples are truncated at the unchanged source/target limits. A visited row is therefore not proof of seeing every token.

The new optional full-coverage path uses CPU-prepared eligible row indices, deterministic per-shard permutations, disjoint chunks mixed across shards, and the existing length grouping. Checkpoints save the cursor after the last successful update, per-shard consumed counts and a row-order hash. Prefetched rows are not counted. Resume checks dataset, tokenizer, seed, batching and optimizer-step identity. Old checkpoints begin a new tracked traversal; prior unique coverage remains unknown.

The real-data CPU row-plan check visits exactly 11,836,444 eligible row IDs, with zero repeated IDs, missing eligible rows or included held-out rows. Its 49 rounds require 92,473 batches at batch size 128, including partial round-end batches. This verifies the planner, not a trained full pass or a score gain. Synthetic CPU tests cover uneven/empty shards, original duplicate rows, partial batches, shuffled grouping, prefetch, every-batch resume, and actual tiny-model training plus archived-checkpoint resume.

The longer 78M run remains on hold. The next GPU action must be a bounded throughput/quality comparison with checkpoint export and greedy bot evaluation. The course review found warm-up, clipping, mixed precision, length grouping, RMS normalization, relative position information, gated feed-forward layers and inference caching already present. Faster attention and alternative gated activations remain unmeasured opportunities; failed short screens do not exhaust their full design space.

Sources: full_coverage.py, full_coverage_preflight.py, tests/test_full_coverage.py; experiments/coverage-readiness-cpu-v1 and full-coverage-cpu-v1. Course material: seminar 2; lecture 3 pp.38-47; lecture 4 pp.6-14,19,35-36,47-52. No GPU was used for the coverage preflight. No public score changed.

# Full-coverage pipeline: short GPU check

Two matched continuations start from the same mature scratch-trained 42.48M model at step 73,024, retaining AdamW moments. Each trains all parameters for 1,024 updates with batch 128, BF16, original data/tokenizer, numerical-token weight 2, 50-update warm-up and peak learning rate 0.00003. Compare the old mixed sampler with deterministic traversal without repeated row IDs. Sampling order differs by design. Both final checkpoints and exact greedy outputs were saved to Drive, then exported, checksum-verified and scored in the course bot.

| Traversal | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| control | 136 | 282 | 40 | 568 |
| coverage | 136 | 291 | 45 | 573 |

The corrected full-coverage path passes the 1,024-update engineering screen: 131,072 consumed original row IDs, no repeats, checkpoint step/cursor agree. Held-out scores are 136/245 and 291/493 versus control 136/245 and 282/493; total execution failures increase from 40 to 45, at the predeclared allowed boundary. Public scores are 573/960 versus 568/960. The previous best remains 575/960. Adopt exact traversal/accounting for future coverage experiments; this short single-seed check does not establish a long-term accuracy improvement or a completed training pass. No long run started.

The driver took 276.22 seconds (4.60 minutes), including restoration, both training arms and evaluation. Training process seconds were 96.69 control and 96.21 coverage. Whole-driver GPU utilization averaged 62.59% with median 96%; peak memory 21465 MiB. These exclude Colab allocation and Drive authorization time. Active-sessions UI confirmed no sessions after completion; its resource sidebar lagged behind.

Steady logged training intervals processed approximately 1,587 examples/s for control and 1,576 examples/s for coverage. A single short comparison cannot establish a small speed difference. The full GPU ledger records 131,072 distinct consumed rows across all 60 shards (1.11% of eligible rows), zero repeated presentations and zero complete passes. Historical exposure before step 73,024 remains unknown. CPU planner verification covers all 11,836,444 eligible rows exactly once; that is separate from actual training exposure.

The predeclared engineering gate rejects a loss greater than 5/245 or 10/493, or more than five extra total execution failures. This run passes at the failure-count boundary; it does not meet a stronger claim of fewer failures or independent replication. Public scoring is recorded separately from the held-out gate. The 78M long run remains on hold. Faster attention and SwiGLU comparisons remain outstanding; no claim of exhaustive optimization is made.

Seventeen focused CPU checks and finite-gradient checks of both full-size paths passed before GPU allocation. Source: 4ad9e1ec3a2991780bc835a65ffe710662cbb3db. Evidence: experiments/coverage-pipeline-1k-v1. Bot messages 788989 and 788994; last reported rank 10/16 at 20:20 on 16 October 2026 (+03). T4 timing is still unverified.

# Attention implementation: bounded speed probe

The course introduces faster attention kernels. This probe keeps the same scratch T5 architecture and original training examples, replacing its attention calculation with PyTorch scaled_dot_product_attention (SDPA). The adapter explicitly preserves T5 scale=1.0, learned relative-position bias and its gradient, padding/causal masks, cross-attention cache and evaluation dropout behavior. Native attention remains the default. Requested attention weights or masked/pruned heads use the original implementation.

Six synthetic CPU tests pass, covering logits, loss, gradients, cache/no-cache greedy generation, deeper decoders, save/reload, diagnostic fallback and dropout settings. A separate full 42.48M CPU test on four original examples passes: maximum logit difference 6.68e-6 and maximum parameter-gradient difference 1.37e-7. Both learned relative-bias tables receive nonzero gradients. These tests establish a limited numerical gate, not a full GPU quality guarantee.

Use 8,192 eligible original shard-0 examples, batch128, BF16, the original 8k tokenizer and a fresh 6/4, width512, feed-forward1536 model. Each trial starts at seed42, performs three warm-up updates, then times the same examples with Muon 0.001 and AdamW 0.0003. Two rounds reverse the native/SDPA order. A separate untimed forward/backward profiler call identifies dispatched kernels. No trained weights are retained.

| Attention | Round 1 examples/s | Round 2 examples/s | Peak allocated GiB |
|---|---|---|---|
| native | 1469.1 | 1503.7 | 15.64 |
| sdpa | 1360.1 | 1352.5 | 16.20 |

Retain native attention. SDPA median 1356.30 examples/s versus native 1486.39 (-8.75%). Both reversed timing rounds agree. This adapter misses the required 5% speed gain and uses more peak allocated memory. No quality pilot or production adoption is justified for this version.

Whole benchmark process 29.35 seconds, excluding allocation, Drive mounting and installation. RTX PRO 6000 Blackwell, BF16, batch128, identical 8192 original examples in each timed run. Profile reports both efficient-attention and math kernels.

Both SDPA profiles contain efficient-attention and ordinary math operations; this is not evidence that all layers used FlashAttention. Dropout random masks differ between implementations. The predeclared gate requires at least a 5% median speed gain before a separate quality pilot. This version fails that gate. No prediction checkpoint was produced, so there is no new bot submission or public-score claim.

Inputs, source, CPU checks and GPU results were saved to Drive and downloaded back. The runtime has a 240-second worker limit and released automatically; active sessions were verified as zero. Source: 906b422. Evidence: experiments/t5-sdpa-probe-v1. References: week04_modern_llms/lecture4.pdf pp.6-9; PyTorch 2.9 scaled_dot_product_attention documentation; Hugging Face Transformers v4.57.6 T5Attention source. Other implementations and longer sequence workloads remain untested.

# Full-corpus token coverage audit

CPU census of every eligible question and program with the original 8k tokenizer, without truncation or padding. Counts include the end token. All 60 Parquet files and eligible indices are checksum-verified. Exclude held-out question copies; retain original duplicate rows. No test answers are read.

| Original text | Mean tokens | Median | 99th percentile | Maximum |
|---|---|---|---|---|
| Question | 46.58 | 43 | 106 | 741 |
| Program | 73.76 | 64 | 219 | 1208 |

| Input / output cap | Questions cut | Programs cut | Either cut (%) |
|---|---|---|---|
| 128 / 192 | 19,870 | 212,045 | 1.927% |
| 192 / 192 | 390 | 212,045 | 1.794% |
| 128 / 256 | 19,870 | 52,192 | 0.597% |
| 128 / 384 | 19,870 | 1,184 | 0.177% |
| 192 / 384 | 390 | 1,184 | 0.013% |
| 256 / 512 | 62 | 153 | 0.002% |

All 11,836,444 eligible rows were audited. Current limits cut text in 228,035 rows; 192/384 would reduce this to 1,551. These are measured counts, not an accuracy improvement.

Test a larger output limit separately from a larger input limit, using short quality and throughput comparisons. Retain original targets, exact row tracking and greedy decoding. Production limits remain unchanged.

Per-row length arrays, per-shard histograms, cap counts, hashes and package versions are retained. Audit resume verifies inputs and array hashes. Three synthetic checks cover quantiles, exclusion/cap accounting, resume and non-string rejection.

The final scan took 700.30 seconds on local CPU. No unknown tokens, literal pad tokens or internal end tokens were found. Larger proposed limits still cut rare outliers: complete token coverage requires accommodating the observed maxima or an explicit outlier path.

A fixed seed selects 32 over-limit original programs per shard (1,920 total). All original programs parse. After applying the current 192-token target truncation, 472 no longer parse, 1,344 still parse but lose every return statement, and only 104 remain parseable with a return. This is an equal-sized per-shard sample conditional on being too long, not a population-weighted estimate. Parsing does not establish execution or numeric correctness. No programs are executed or rewritten.

A corpus census does not prove training exposure. The best checkpoint still lacks a completed full-pass receipt. Exact traversal now prevents repeats before a full pass and saves its consumed position, but the short pipeline check is not a full epoch. No optimal context limit, accuracy gain or exhaustive optimization is claimed.

Source: corpus_token_audit.py, tests/test_corpus_token_audit.py, commit 906b422. Evidence: experiments/corpus-token-cpu-v1. Full per-row arrays, source and summary were uploaded to Drive and verified by download. The existing best inference artifact and its public score remain unchanged.

# Retaining longer original training programs

The corpus audit found 212,045 programs longer than the current 192-token training cap. Compare caps 192 and 384 in two 1,024-update continuations from the same own scratch-trained 42.48M parent at step 73,024. Keep AdamW moments, batch 128, BF16, seed 42, rate 0.00003, warmup 50 and numerical-token weight 2. All parameters train. Both arms use the corrected full-coverage traversal.

Grouping uses target lengths capped at 192 in both arms, preserving identical row order and batch membership. Only the retained training target length changes. Source cap 128, validation target cap 192 and greedy generation cap 192 stay fixed. No code rewriting or extra labels are used.

| Training cap | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 192 | 135 | 291 | 45 | 573 |
| 384 | 138 | 284 | 48 | 570 |

The 384-token training cap fails the predeclared short-run quality gate: held-out correct-answer changes are +3/245 and -7/493 versus the 192-token control; execution failures change by +3. Public scores are 570/960 versus 573/960. The previous best remains 575/960. Do not combine this change on the basis of this short trial. Retaining complete original targets remains a data-integrity requirement to address, and a short negative continuation does not rule out a longer-run benefit. No long run started.

The driver took 296.55 seconds (4.94 minutes), including restoration, both training arms and evaluation. Training process times were 98.28 and 116.56 seconds. Steady logged throughput was 1567.3 versus 1270.1 examples/s. Whole-driver GPU utilization averaged 65.69% (median 96%); peak memory was 46411 MiB. Allocation, Drive sign-in and installation are outside this driver timing.

The predeclared selection gate requires positive executed-answer gains on both held-out subsets, no extra total execution failures and no loss against the parent on either subset. Any survivor needs a second seed. Public scores are recorded separately. These reused held-out sets and short continuations cannot establish statistical significance or the best long-run context size.

Both checkpoints record exactly 131,072 unique original rows, zero repeats and the same consumed-row-order hash. This is about 1.11% of the eligible corpus, not a complete training pass. The 384-token limit still truncates 1,184 eligible programs; rare long questions and programs need explicit handling before claiming complete token coverage. The unchanged generation cap can limit the benefit on long outputs.

Eleven CPU tests passed, including default-stream compatibility, changed-cap resume rejection, matched row order and actual checkpoint/archive/resume. On 4,096 original rows, 80 targets retain additional text; target tokens rise from 300,695 to 304,642 while padded positions rise from 720,896 to 912,512. A full-size parent forward/backward check on longer original programs has finite gradients. No CPU-test weights were saved.

Source 698b8bf1520b93904919c2acb07e2c2102341774. Every retained checkpoint and strict greedy output was saved to Drive before export, checksum verification and bot evaluation. Evidence: experiments/target-length-1k-v1. No long run started; T4 timing remains unverified.

# SwiGLU versus gated GELU from scratch

Two fresh 42,481,152-parameter encoder-decoder Transformers differ only in the gated feed-forward activation: T5 gated GELU versus SiLU gating (SwiGLU). Both use six encoder layers, four decoder layers, width 512, feed-forward width 1536 and eight attention heads. Full initial parameter hashes, including embeddings, agree. The activation is stored in the model configuration and is verified after reload.

Train all parameters for 2,048 updates with seed 42, batch 128, BF16, warmup 200 and a cosine schedule. Native Muon uses rate 0.001 on hidden matrices; AdamW uses 0.0003 on embeddings and vectors. Keep the original tokenizer, original examples/targets, source/target caps 128/192, ordinary token loss and strict greedy decoding. Record validation loss every 512 updates and retain only the final checkpoint from each arm.

| Activation | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| Gated GELU | 26 | 49 | 159 | 107 |
| SwiGLU | 20 | 53 | 191 | 108 |

SwiGLU fails the predeclared early-quality gate: held-out gains are -6/245 and +4/493, with +32 execution failures versus gated GELU. Public scores are 108/960 versus 107/960. Keep gated GELU; no second-seed quality confirmation or long run is triggered. Faster short training is a useful observation, but this unequal-time test does not establish accuracy at matched compute. The retained mature best stays 575/960; current public rank is 10/16 at 22:47 (+03:00).

Driver duration 447.17 seconds (7.45 minutes), including restoration, both arms and evaluation. Training-process times were 209.49 and 166.92 seconds. Logged training throughput was 1339.9 versus 1701.9 examples/s. Whole-driver GPU utilization averaged 69.58% (median 90%); peak memory 18681 MiB. Allocation, Drive sign-in and installation are outside driver timing.

The predeclared gate requires positive executed-answer gains on both held-out subsets and no additional total execution failures. Any survivor needs a second seed before combination or extension. Public feedback is recorded separately. These short scratch models are compared with each other, not interpreted as mature replacements for the 73,024-step best checkpoint. Equal updates are not equal GPU time; activation-specific learning rates were not tuned.

Checkpoint ledgers verify the same 262,144 unique original rows in the same order, zero repeats and zero completed passes. Coverage is about 2.21% of the eligible corpus. Existing sequence truncation is deliberately held fixed to isolate the activation change; this test does not resolve complete-token coverage or establish an optimal final model.

Ten focused CPU tests passed: parameter counts, identical initialization, distinct activation functions, finite gradients, save/reload, cached/uncached greedy decoding, resume-override rejection, command construction and selection/record checks. Separate full-size CPU forward/backward and native Muon updates on four original examples are finite for both variants; no preflight weights were retained. All 60 data shards and eligible indices were checksum-verified before allocation.

Source 2735e1e29779849bdaa359c92d2b62f75650dd36. Evidence: experiments/activation-muon-2k-v1. Checkpoints and outputs were saved to Drive, exported and checksum-verified before bot evaluation. Research motivation: Noam Shazeer, GLU Variants Improve Transformer, https://arxiv.org/abs/2002.05202. Its findings are motivation, not evidence for this task. No long run started; final T4 timing remains unverified.

# Complete original sequences before scaling

All 11,836,444 eligible original rows are indexed, but no competition checkpoint has a verified full training pass. A complete pass requires 92,473 logical batches at 128. Independent pilot models cannot combine their training exposure. Keep the longer 78M run on hold.

The new opt-in --untruncated-training path keeps complete questions and programs. Grouping retains the old capped-length ordering to isolate token retention. Long logical batches split on CPU before GPU transfer. Each chunk receives its share of the original loss denominator, including numeric-token weights; rows and complete-token counts are committed only after the optimizer update succeeds.

| CPU check | Observed result |
|---|---|
| Eligible row lengths checked | 11,836,444 |
| Longest question / program | 741 / 1,208 tokens |
| Selected long original examples | 130; exact token IDs retained |
| Original mixed window | 4,096 rows; unchanged row order |
| Logical batches / GPU chunks | 32 / 44; GPU execution not performed |
| Question / program tokens in window | 193,287 / 304,642 |
| Focused CPU tests | 24 passed; checkpoint and resume included |

Default chunk limits are 40,960 padded source-plus-target tokens and 9,961,472 attention positions (batch times source squared plus target squared plus source times target). All recorded individual examples fit. These are proxies, not measured device memory. An oversized single example raises before any chunk is yielded; no row is silently cut or dropped.

Dropout-free tiny T5 gradient comparisons verify split/unsplit loss equivalence for ordinary, numeric-weighted, example-balanced, smoothed and focal loss. Dropout and floating-point rounding can still differ in real training. Resume rejects a changed truncation mode. No test weights are used as a competition model.

The fresh 42.48M gated-GELU versus SwiGLU screen completed after CPU-only Drive recovery. Its results are reported on the preceding page. The earlier failed mount started no training worker. Recovery verified source and index hashes and a durable Drive read/write receipt before the successful GPU launch.

Next: run a bounded full-sequence memory/throughput probe and confirm any promising optimization before combination. Save every retained checkpoint and greedy output to Drive before export and bot scoring. Require a full consumed-data receipt before claiming all examples were trained. Optimization search is prioritized and finite, never described as exhaustive.

Evidence: experiments/untruncated-cpu-v1 and experiments/activation-muon-2k-v1/cpu-recovery-status.json. The whole-corpus length arrays were hash-verified; selected original extremes were retokenized. Validation and generation caps remain fixed, so this work does not establish complete evaluation context or improved accuracy.

# Complete-sequence GPU feasibility verified

The CPU-prepared source a88585879408275e8060b64d851ca5b8b74e3a11 ran on an NVIDIA RTX PRO 6000 Blackwell Server Edition with PyTorch 2.11.0+cu130. Each variant used a fresh 42.48M 6/4 model, seed 42, BF16, Muon 0.001 / AdamW 0.0003 and numeric-token weight 2. All temporary weights were discarded.

Compare the same 4,096 original examples in capped/complete and complete/capped order. Each run has three warmup updates followed by 32 measured updates. Complete batches retain the verified 193,287 question and 304,642 program tokens; they split into 44 GPU chunks while preserving logical batches and loss normalization.

| Order / variant | Examples/s | Peak allocated GiB | GPU chunks |
|---|---|---|---|
| 1 / capped | 1,365.1 | 17.54 | 32 |
| 1 / complete | 1,160.5 | 16.93 | 44 |
| 2 / complete | 1,165.0 | 16.93 | 44 |
| 2 / capped | 1,379.2 | 17.54 | 32 |

Complete sequences average 1,162.7 examples/s versus 1,372.2 capped: 18.0% more time for the same examples. Complete peak allocated memory is 16.93 GiB; the allocator reserves up to 24.72 GiB. This is not a whole-corpus throughput estimate.

The separate stress test executed all 130 selected original length extremes in 15 chunks, retained every token, produced finite loss, and peaked at 8.22 GiB allocated memory. Every optimizer update checks finite gradients. This tests selected extremes, not every original example on GPU; the complete-corpus length feasibility check was performed on CPU.

Measured driver time was 18.05s and the worker lifetime was 27.38s. The 91 telemetry samples have median utilization 95% and mean 75.35%, including initialization and cleanup. Allocation, Drive authorization and installation are additional. Worker exit 0 and zero active Colab sessions were verified; results were exported from Drive.

A later CPU-only chunk-count check found 44/34/32/32 ordinary chunks at 1x/2x/3x/4x budgets, and 15/9/7/6 extreme chunks. Larger budgets have not been measured on GPU. They may reduce launch overhead but require separate memory and speed evidence before adoption.

Next: choose a measured chunk budget, then run a bounded accuracy continuation with exact consumed-row/token receipts and multiple Drive-saved, bot-scored checkpoints. No accuracy gain, completed training pass, optimal model size or final T4 performance is claimed. Scaling remains on hold.

Evidence: experiments/untruncated-gpu-probe-v1/results.json, conclusion.json, gpu.csv, execution.complete.json and drive-results.json. The preceding CPU-only section records the state before this GPU check. No competition checkpoint was created by this probe.

# Bounded complete-sequence accuracy curve

Continue our own 42.48M best checkpoint from step 73,024 to 76,800: 3,776 updates, with four retained checkpoints. Keep AdamW moments, BF16, batch 128, numeric-token weight 2, peak learning rate 0.00003 and 50 warmup updates. Complete original questions and programs use the GPU-verified conservative chunk budgets; no larger model or new targets are introduced.

Parent held-out agreements were 139/245 and 286/493, with 40 execution failures; public score was 575/960. The table reports all retained checkpoints. Their exact original row and full-token counts are committed only after successful optimizer updates. Old historical exposure remains unknown.

| Step | Unique rows | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|---|
| 73728 | 90,112 | 139 | 285 | 38 | 574 |
| 74752 | 221,184 | 136 | 295 | 42 | 581 |
| 75776 | 352,256 | 139 | 291 | 43 | 583 |
| 76800 | 483,328 | 140 | 288 | 43 | 569 |

The public trajectory is 574, 581, 583 and 569 correct out of 960. Retain step 75,776 at 583/960 (60.73%) as the new public best, an 8-answer gain over 575/960; latest rank is 10/16 at 23:15 (+03). Held-out gains for this checkpoint are 0/245 and +5/493, with 3 additional execution failures. The final checkpoint gains 1/245 and 2/493 but falls publicly. This is mixed evidence, not a repeatable quality gain or support for an automatic long extension. The selected checkpoint has 352,256 newly tracked unique rows; the final one has 483,328. Neither completed a full pass.

The driver took 512.01 seconds (8.53 minutes), including restoration, training, checkpoint exports and held-out evaluation. Training-process time was 408.97 seconds. Logged throughput was 1,232.4 examples/s, including intermediate checkpoint work. Whole-driver GPU utilization averaged 71.28% (median 94%); peak memory was 34,633 MiB. Allocation, Drive authorization and installation are additional. Worker exit 0 and zero active sessions were verified.

This is a short continuation trajectory, not a matched causal comparison of truncation or data traversal. Held-out sets have been reused. Strict greedy decoding and the existing 128/192 validation/generation limits stay fixed for comparability. Public feedback is recorded separately; a score difference alone does not establish a repeatable training improvement.

Twenty-six CPU tests passed, covering complete token retention, loss normalization across chunks, exact resume, checkpoint scheduling and rejection of inconsistent coverage receipts. Before GPU allocation, parent weights and AdamW moments were verified; the actual parent produced finite forward/backward values on original examples reaching 741 question and 1,208 program tokens. All 60 data/index shards were verified.

Every retained checkpoint and its 1,919 model-generated programs were saved to Drive, exported with model/output hashes verified, and evaluated by the course bot. No handwritten fixes, beam search or target augmentation were used. This run does not complete a full data pass and does not establish final T4 inference timing.

Source d3c8136a68830bca45f26ab4fcad31450d344f81. Evidence: experiments/complete-curve-4k-v1, per-checkpoint coverage and bot receipts, training logs, GPU telemetry, and execution/runtime-release records.

# Adjacent checkpoint averaging: rejected

A CPU-only paired audit of the complete-sequence curve found that step 75,776 gained 16 held-out answers and lost 17 relative to step 74,752. This motivated two fixed arithmetic means: steps 74,752/75,776 and steps 74,752/75,776/76,800. The baseline was step 74,752, selected for its highest aggregate saved held-out accuracy rather than public score.

All three original model hashes were checked against their Drive checkpoint manifests. Both averaged models were saved to Drive, downloaded and byte-verified, then evaluated on CPU with FP32 and strict greedy decoding. No training updates, new targets or manual output changes were used. Arithmetic averaging does not merge data-coverage receipts, and these inference-only weights have no valid inherited optimizer state.

| Model | Pilot /245 | Confirm /493 | Failures |
|---|---|---|---|
| baseline | 136 | 295 | 42 |
| avg74_75 | 139 | 293 | 43 |
| avg74_75_76 | 140 | 289 | 42 |

Both candidates improved the pilot subset but declined on the separate confirmation subset. The two-model mean also adds an execution failure. Neither passes the predeclared requirement of improvement on both subsets with no added total failures. No candidate is exported to the public test or adopted for training.

The existing public best remains 583/960 (60.73%) from the trained step 75,776 checkpoint. The held-out subsets have been reused, so results are exploratory; this rejects these fixed averages, not all possible averaging methods.

Three averaging tests passed, including exact means, immutable parents and incompatible-tokenizer rejection. Source d811ba5. Evidence: experiments/adjacent-average-cpu-v1, paired audit in experiments/scaling-release-v1/checkpoint-stability.json, and Drive archive verification receipts.

# Complete-sequence batch budget comparison

Compare 1x, 2x and 3x memory-budget proxies at the same effective batch of 128 and the same original examples. The 42.48M random model uses AdamW at 0.00003, BF16, gradient clipping and numeric-token weight 2. Each run has three warmup and 32 measured updates; two reversed orders reduce order effects. Weights are discarded.

| Repeat / budget | Examples/s | Allocated GiB | Chunks |
|---|---|---|---|
| 1 / 1x | 1,295.8 | 17.07 | 44 |
| 1 / 2x | 1,207.8 | 24.78 | 34 |
| 1 / 3x | 1,198.5 | 29.61 | 32 |
| 2 / 3x | 1,199.1 | 29.61 | 32 |
| 2 / 2x | 1,212.6 | 24.78 | 34 |
| 2 / 1x | 1,310.4 | 17.07 | 44 |

Retain the 1x budget. Mean throughput is 1,303.1 examples/s versus 1,210.2 at 2x and 1,198.8 at 3x. Larger budgets take 7.67% and 8.69% more time for the same examples, with greater memory use. Both reversed-order repeats reject a speed improvement. All regular and 130-example extreme tests completed with finite updates; fewer GPU chunks did not improve throughput.

The measured driver took 26.70 seconds; worker lifetime was 36.23 seconds. Across 133 GPU samples, median utilization was 98.0% and mean 84.41%; peak sampled memory was 37,669 MiB. Allocation, Drive authorization and installation are additional. Worker exit 0 and zero active Colab sessions were verified.

Each budget also processes 130 original length extremes. CPU checks preserve every non-padding question and program token in order, and GPU updates check finite gradients and loss. The effective batch, consumed examples and loss normalization stay fixed. Splitting can change dropout draws and floating-point rounding; speed measurements are not accuracy proof.

The predeclared adoption rule requires a speed gain in both reversed-order repeats and at least 15% device-memory headroom. Timings cover pretokenized batches, transfers, forward/backward, clipping and optimizer updates. Tokenization, allocation, sign-in and installation are additional; this is not a full-corpus training-time estimate.

Source a4c7e96. Evidence: experiments/complete-budget-gpu-v1, CPU preflight, runtime completion, Drive exports and telemetry. No new competition checkpoint or full-pass claim results from this probe.

# Second-seed model-size confirmation

Repeat the shortlisted 17.58M versus 42.48M scratch-model comparison with seed 1042 and reversed run order. Keep the original tokenizer, Muon rates, batch 128, 2,048 updates and capped mixed-sampler settings from the first size screen. This isolates replication; it does not combine later full-coverage or complete-sequence changes.

| Seed / model | Pilot /245 | Confirm /493 | Failures | Public /960 |
|---|---|---|---|---|
| 42 / control | 22 | 37 | 219 | 107 |
| 42 / small | 27 | 49 | 117 | 90 |
| 1042 / control | 21 | 51 | 165 | 114 |
| 1042 / small | 20 | 43 | 152 | 103 |

The smaller model does not replicate its initial gain: seed42 gains were +5/245 and +12/493; seed1042 gains are -1/245 and -8/493. Reject promotion of the 17.58M candidate and retain 42.48M as the reference, not a proven optimum. The short public scores are 114/960 for control and 103/960 for small; the mature 583/960 checkpoint remains best. Proceed to the bounded matched-compute learning-curve comparison before the authorized full-pass run.

The two-arm driver took 407.92 seconds (6.80 minutes), including restoration and evaluation. Training process times were control 202.35s and small 136.73s. Each arm consumed 262,136 examples in 2,048 updates; partial batches explain the difference from the nominal 262,144. Whole-driver GPU utilization mean 69.91%, median 92%; peak memory 18309 MiB. Both retained checkpoints were saved to Drive, exported with verified hashes and bot-scored. Zero active Colab sessions was verified after completion.

The predeclared rule requires positive gains on both held-out subsets and no extra total execution failures in both seeds. Equal updates match example exposure, not GPU time. Two early learning seeds do not establish an optimal final model size. Public scores are recorded separately and do not select the recipe.

Eleven CPU tests passed. Actual-width models had finite gradients, unchanged original labels, and matching greedy outputs with and without decoding cache. Both final checkpoints were saved to Drive, downloaded with model hashes verified, and their 1,919 strict-greedy outputs were scored by the course bot.

Source f2d3b16. Evidence: experiments/size-confirm-2k-v1, prior selection hash, model/output verification, GPU telemetry and bot receipts. The longer run remains conditional on the bounded matched-compute comparison using complete original sequences and exact consumed-row tracking.

# Fresh question groups for the longer comparison

The original small held-out subsets have been reused across many experiments. Reserve 2,048 distinct original question groups from shard 59 using a fixed SHA-256 ordering with seed 20261017. Selection uses question text only; retain the first original program for each question without changing it. Exclude every question text present in the question column of shard zero, which also supplied tokenizer fitting. Text embedded inside programs/comments was not audited for overlap.

All 60 original shards were hash-checked. Remove every occurrence of each reserved question from the existing eligible-row index: 2,583 additional rows are excluded, leaving 11,833,861 eligible training rows. This also preserves all previous held-out exclusions. The derived index and original question/program file were saved to Drive, downloaded and byte-verified before reference diagnostics.

| Subset | Reserved questions | Usable references |
|---|---|---|
| pilot | 1024 | 928 |
| confirmation | 1024 | 943 |

The restricted numeric runner reports NameError or ValueError for some supplied programs. These are evaluation limitations, not proof that those references are wrong. No questions, programs or split assignments were changed after observing these statuses. Report accuracy with the usable-reference denominator and report generated-program failures separately.

This is a clean model-training holdout only for future scratch models that use the derived exclusion index. Historical checkpoints may already have trained on these questions and must not be described as independently evaluated on unseen data. The first 1,024 reserved positions form the pilot subset; the remaining 1,024 are the confirmation subset.

At logical batch 128, accumulation 1 and 4,096-row shard chunks, one exact pass needs 92,453 optimizer updates across 49 rounds, including 2 partial batches. This is a calculated budget, not a completed training pass. The later long-run schedule must use the complete-pass budget instead of repeatedly restarting a short cosine cycle.

Four CPU tests passed for answer-independent deterministic group selection, duplicate exclusion, preserved original program strings and ordered explicit evaluation inputs. Source 9ca08e1. Evidence: experiments/fresh-holdout-v1 and the verified Drive archive; no model accuracy claim is made from the reference audit.

# Final matched-time comparison: preparation

The smaller model failed its second-seed confirmation. The remaining finite comparison tests optimizer and larger capacity at a longer early-learning horizon with equal measured training time. This revisits 78M for a different budget question; the earlier 2,048-update result did not establish its later learning curve. This page documents CPU preparation, not measured model accuracy.

| Candidate | Parameters | Measured training | Retained checkpoints |
|---|---|---|---|
| AdamW reference | 42,481,152 | 600 seconds | 300s and 600s |
| Muon, same size | 42,481,152 | 600 seconds | 300s and 600s |
| Muon, larger size | 78,146,112 | 600 seconds | 300s and 600s |

Each model starts from random weights and uses the same original 8K tokenizer, 6 encoder / 4 decoder layers, complete original questions and programs, effective batch 128, BF16, and unweighted token loss. The 42M models have identical initial parameter hashes. Keep the previously measured conservative microbatch budgets. Training time includes input waits and complete optimizer updates; checkpoint, test generation and evaluation overhead is reported separately.

Replace the unused single-shard holdout before model evaluation: reserve 2,048 question groups, 34 or 35 from each of shards 1 through 59, then mix their order by a fixed question-text hash. Exclude all tokenizer-shard question texts and every reserved question copy from training. This removes 2,221 rows from the original eligible index, leaving 11,834,223. Tokenizer overlap inside program text remains unaudited; historical models are not clean controls on this split.

The unchanged restricted runner accepts 982/1,024 pilot references and 988/1,024 confirmation references. No selection or target rewriting uses those results. A model candidate must improve both subsets without added total execution failures. Repeat a winning alternative against AdamW42 with another seed before the authorized full pass; public scores are reporting only.

The trainer now stops at a completed update and saves its actual step and measured training time. Checkpoints retain model, optimizer, scheduler, consumed-row cursor and random-number states. A CPU integration test verifies that interrupted/resumed training produces exactly the same weights as uninterrupted training. The learning-rate schedule spans the calculated 92,456-update full pass with 500 warmup updates, so the short comparison does not force a premature cosine decay.

Greedy evaluation and test export now support complete questions. The new comparison sets a 1,280-token generation limit, above the audited 1,208-token longest original target, and records limit hits. A focused test checks input retention, original output order and correct limit counting when shorter finished outputs are padded. No handwritten corrections or beam search are introduced.

Thirty-nine focused CPU tests passed; six final timing/selection tests also passed after adding per-checkpoint timing. Actual 42M AdamW, 42M Muon and 78M Muon models had finite updates and identical greedy outputs with and without decoding cache. CPU Muon update timings are not GPU speed estimates. The data bundle was saved to Drive, downloaded and byte-verified.

Source 723f5b3. Evidence: experiments/matched-curve-v1 and experiments/fresh-holdout-v2. The current public best remains 583/960; no score improvement or completed full pass is claimed from preparation.

# Matched-time comparison: measured results

All models are trained from scratch under the previously stated common recipe. The fixed reference subsets contain 1,024 rows each; numeric agreement uses 982 and 988 usable original references. Counts below are executed-answer agreements, not token accuracy. Training time excludes checkpoint/export/evaluation work.

| Model | Updates | Training s | Pilot /982 | Confirm /988 | Public /960 |
|---|---|---|---|---|---|
| adamw42 | 6,018 | 600.0 | 122 | 136 | 158 |
| muon42 | 5,422 | 600.0 | 107 | 129 | 134 |
| muon78 | 3,871 | 600.0 | 40 | 38 | 55 |

At 600 training seconds, Muon42 changes correct answers by -15 / -7 versus AdamW42, with -50 total generated-program execution failures. It fails the promotion rule because both agreement counts decline. Fewer failures alone do not establish better answers.

The predeclared rule selects adamw42. Neither alternative passes; retain AdamW42 for the authorized full pass without another confirmation run.

## Retained checkpoint exports and public bot receipts

| Checkpoint | Training s | Public correct | Public score |
|---|---|---|---|
| adamw42 / 3,011 | 300.0 | 98 | 10.21% |
| adamw42 / 6,018 | 600.0 | 158 | 16.46% |
| muon42 / 2,686 | 300.3 | 82 | 8.54% |
| muon42 / 5,422 | 600.0 | 134 | 13.96% |
| muon78 / 3,871 | 600.0 | 55 | 5.73% |

Every listed submission was first saved on Drive, downloaded, checked against its model hash and selected token IDs, and sent once in official test.parquet order. Strict greedy decoding and original model outputs are preserved. Public feedback is reported separately from recipe selection. Different models consume different row counts within equal measured time.

Live leaderboard evidence checked 2026-10-17T00:44 (+03): leader 78.33%; our retained mature checkpoint 60.73%, rank 10/16. First place is not achieved. These early scratch checkpoints do not replace the retained best.

Source 723f5b3 for the live comparison. Evidence: experiments/matched-curve-v1, exact model/output hashes, separate bot receipts, and leaderboard-live.json/png. The longer-run driver preserves the selected seed 42 optimizer, schedule, RNG and consumed-row position, verifies every eligible row exactly once at update 92,456, and measures final fixed-subset agreement. No longer-run result is inferred from the driver checks.

Export exception: the 78M step 1,921 output exceeded the bot limit (4.88 MB versus 4 MB). Original predictions remain on Drive. A CPU size estimate selected a 640-token cap for fresh greedy generation from the same weights; no programs are edited. The regenerated file scored 49/960 (5.10%). This export uses a different cap; held-out comparisons retain 1,280 tokens. See bot_rejection.json, reexport-size-audit.json and the nested export receipt.

# Selected recipe: full-data scale-up

The completed matched-time comparison selected AdamW42. Continue its original seed-42 checkpoint at update 6,018 to 92,456, preserving optimizer moments, learning-rate schedule, random states and the consumed-row cursor. The planned pass covers 11,834,223 eligible original rows; validation groups stay excluded.

Keep the 42,481,152-parameter encoder-decoder, original tokenizer and targets, BF16 training, effective batch 128, length grouping and two-batch prefetch. Microbatches are bounded by token and attention budgets while preserving complete input and target sequences. Public exports use FP32 greedy decoding, complete questions and a 1,280-token output cap.

Launch v1 stopped before training because fresh-model flags were passed to checkpoint resume; its GPU guard released the runtime. Removing those flags passed ten CPU checks and a one-update CPU resume smoke test. Production resumed the unchanged GPU checkpoint; FP32 and skipped CUDA RNG restoration applied only to the CPU test.

Full-pass coverage has been verified against the final saved ledger.

## Drive-verified checkpoint submissions

| Update | Unique eligible rows seen | Public correct /960 | Public score |
|---|---|---|---|
| 10,000 | 1,280,000 | 221 | 23.02% |
| 20,000 | 2,559,950 | 369 | 38.44% |
| 30,000 | 3,839,950 | 458 | 47.71% |
| 40,000 | 5,119,950 | 508 | 52.92% |
| 50,000 | 6,399,950 | 533 | 55.52% |
| 60,000 | 7,679,950 | 560 | 58.33% |
| 70,000 | 8,959,950 | 581 | 60.52% |
| 80,000 | 10,239,950 | 579 | 60.31% |
| 90,000 | 11,519,950 | 591 | 61.56% |
| 92,456 | 11,834,223 | 592 | 61.67% |

Independent CPU reconstruction matches the saved consumed-row hash chain, per-shard counts and full-token totals at updates 6,018, 10,000, 20,000, 30,000, 40,000, 50,000, 60,000, 70,000, 80,000, 90,000, 92,456. It used verified original row IDs and cached token lengths, without importing the production sampler or allocating a GPU. The complete plan contains 11,834,223 distinct eligible rows with no repeats. Uncompared future checkpoints are planned coverage, not proof of completed training.

Individual regression: the user-identified triangle question returned 4.5 at step 40,000, while the correct height is 9; step 20,000 returned 9.0. The overall score improved, but individual answers are not monotonic. This observation is retained without output repairs or training-target changes.

At step 50,000 the unchanged generated triangle program again uses (2 * area) / base, which gives 9 for area 54 and base 12. This is a source/arithmetic inspection, not an additional execution benchmark. At step 60,000 the bot again reports 4.5 instead of 9.

Each checkpoint is saved to Drive before its exact token IDs, CSV and 1,919-row JSONL are downloaded and checked. Bot receipts remain separate from training and held-out measurements. Saving and inference overhead are excluded from training-time estimates. The runtime has a hard timeout and automatic GPU release on completion or failure.

Frozen training source 82af821; evidence: experiments/selected-full-pass-v1 (failed launch) and selected-full-pass-v2. Historical public best remains a separate artifact; neither one completed pass nor a green execution receipt alone proves first place or T4 timing compliance.

# Full-pass result and retained model

Retain the 92,456-step checkpoint as the highest observed public score, 592/960 (61.67%), rank 9/16. The bounded full pass is complete. Do not automatically extend training based on the one-answer gain over 90k. First place and T4 timing remain unverified.

The final saved checkpoint independently matches all 11,834,223 eligible original rows, zero repetitions, the exact consumed-row order, per-shard counts and full-token totals. Validation question groups remain excluded. All ten scheduled checkpoints were saved to Drive, downloaded for verification and bot-scored.

## Fixed held-out diagnostics

| Subset | Parent 6,018 | Final 92,456 | Usable references | Failures: parent / final |
|---|---|---|---|---|
| pilot | 122 | 615 | 982 | 99 / 44 |
| confirmation | 136 | 600 | 988 | 110 / 48 |

Fixed unseen question groups with supplied references; agreement is not bot accuracy and reference errors remain possible. Comparison is against the 6,018-step parent, not an equal-compute alternative.

The completed driver took 2.47 hours. Across 8,852 one-second telemetry samples, mean GPU utilization was 87.26%, median 95%, and peak sampled memory 58.79 GiB. The controller exited with code zero; Colab Manage Sessions showed no active sessions after automatic release.

The 90,000-step checkpoint scored 591/960; the final checkpoint scored 592/960. The final retained score improves by nine answers over the historical 583/960 best, but one additional answer over 90,000 steps does not establish a benefit from another long run. Public score selects the retained checkpoint; the recipe was selected before this run using held-out comparisons. No public answers were used as training targets.

The user-identified triangle program at step 90,000 returned 9 in independent restricted execution. At the final checkpoint the bot again reported 4.5 instead of 9. The original generated outputs remain unchanged; aggregate improvement does not guarantee every answer improves.

The retained inference notebook now pins the final checkpoint, frozen source, complete questions, output cap 1,280, batch 64, FP32 and strict greedy decoding. Its code passes syntax checks, but the notebook has not yet been executed end to end. Exact output reproduction and the one-T4 180-second requirement remain unverified. Public rank 9/16 is not first place.

# Post-pass averaging and final size screen

Test two fixed uniform weight averages without more training: steps 90,000 + 92,456, and steps 80,000 + 90,000 + 92,456. Regenerate the final-checkpoint baseline on the same CPU path. The pilot contains 512 fixed held-out questions, with 489 executable original references.

| Model | Matches /489 | Prediction failures | Generation seconds |
|---|---|---|---|
| baseline | 311 | 27 | 418.97 |
| last2 | 307 | 22 | 389.80 |
| last3 | 305 | 27 | 448.51 |

Both averages lose numeric agreement, so neither reaches confirmation or public submission. Fewer execution failures for the last-two average do not offset its four-answer loss. Retain the original final checkpoint. These reused held-out questions make the screen exploratory; no public answers select the averaging weights.

All paths use complete questions, a 1,280-token output cap, batch 64, FP32 and greedy decoding with four CPU threads. Baseline CPU programs match all 512 previously generated GPU programs exactly. No GPU was allocated for averaging. Model archives, outputs and selection were saved to Drive and downloaded for byte verification.

Parent tokenizer vocabularies and special tokens match. The final checkpoint differs only in persisted backend padding/truncation settings. Full-length token IDs match for all 4,096 held-out question/program strings and a long padded probe. The explicit compatibility exception preserves parent assets and rejects vocabulary changes.

## Remaining size question

The corrected matched-time comparison tested AdamW42, Muon42 and Muon78. It did not test AdamW78, leaving optimizer and size incompletely separated. The final short screen adds one scratch AdamW78 arm with 600 measured training seconds and checkpoints at 300 and 600 seconds. Reuse the verified AdamW42 reference to save ten GPU minutes; an earlier runtime remains a host-load limitation.

Require improvement on both 1,024-question held-out subsets without added total prediction failures, with matching question, evaluator, runtime and decoding settings. A provisional larger-model winner needs a fresh paired second-seed confirmation. CPU forward/backward, finite-update, cached-decoding and complete-first-batch checks passed; seven driver tests passed. This preparation does not establish a quality gain.

The user authorized scaling the best recipe after these tests. Close this finite screen, freeze the selected settings, set an explicit longer-run budget and validation stopping rule, preserve complete data traversal and resumable state, save each checkpoint to Drive, and bot-score its unchanged greedy output. Do not claim all possible optimizations have been exhausted.

Evidence: experiments/full-pass-average-cpu-v1 and adamw-size-curve-v1. Size-screen frozen source 58eb536. Current retained public score remains 592/960 (61.67%); first place and one-T4 timing remain unverified.

# Final AdamW size comparison

Complete the missing optimizer-controlled size comparison: 78,146,112 versus 42,481,152 parameters, both AdamW. One fresh 78M run saves five- and ten-minute checkpoints. Keep the original tokenizer, 6 encoder / 4 decoder depth, complete sequences, seed 42, BF16, logical batch 128, full-pass schedule and greedy evaluator. Different sizes process different row counts within equal measured training time.

| 600-second model | Updates | Pilot /982 | Confirm /988 | Public /960 |
|---|---|---|---|---|
| AdamW42 (archived) | 6018 | 122 | 136 | 158 |
| AdamW78 | 4256 | 98 | 93 | 119 |

Final selection: adamw42. The larger model fails the fixed gate. Skip the conditional second-seed GPU run and close the size screen.

The five-minute AdamW78 checkpoint (step 2,109) scored 62/960 (6.46%) publicly; its archived AdamW42 counterpart scored 98/960 (10.21%). The ten-minute AdamW78 checkpoint (step 4,256) scored 119/960 (12.40%), versus 158/960 (16.46%) for AdamW42. Public results report checkpoint behavior; held-out agreement determines the recipe. Our retained mature model stays 592/960 (61.67%), rank 9/16, checked again at 04:52 +03 on 17 October.

Both exact 1,919-question exports were downloaded from Drive, checked against downloaded model weights, CSV strings and selected token IDs, then sent once. Strict greedy decoding uses full questions, FP32, batch 64 and output cap 1,280. Generation took 126.34 and 126.11 seconds on RTX PRO 6000; these measurements do not prove the one-T4 limit. Receipts and verification files were uploaded and downloaded byte-for-byte.

The first-seed 42M baseline is reused from an earlier runtime, saving ten GPU minutes. GPU model, software versions, training schedule, question hashes and evaluator settings are checked. Host-load differences remain possible, and this short comparison does not prove which size would win at every longer budget.

## Preparing the authorized scale-up

The prepared controller trains in resumable intervals of at most 1,000 measured seconds, saves each checkpoint to Drive, and evaluates both fixed held-out subsets. A qualifying improvement must increase both agreement counts without added total execution failures. Stop after three consecutive non-improving intervals, the complete-pass target, or the explicit training-time bound. Every saved checkpoint remains eligible for unchanged-output bot reporting.

If 42M remains selected, continue the completed first pass toward a second pass with a new cosine phase starting at its final learning rate, 3e-5, decaying to 3e-6. Preserve optimizer moments and random/data state. If 78M wins and confirms, preserve its original schedule while finishing its first pass. No alternative branch is launched merely because its code exists.

Six focused controller tests passed, alongside the existing continuation/budget tests. Two real CPU updates from the mature 42M parent verified optimizer steps 92,457 and 92,458, new schedule phase steps 1 and 2, and data-cursor advances of 128 rows each. Pinning NumPy 2.1.3 resolved the expected stream-fingerprint rejection from local NumPy 1.26.4. Parent files were unchanged. CPU FP32 and skipped CUDA RNG restoration apply only to this smoke test; they are not a GPU quality result.

Evidence: experiments/adamw-size-curve-v1, selected-scale-v1 and the separate bot receipts. Size-run source 58eb536. Completion of the leaderboard goal and T4 timing remain unproven.

# Selected 42M model: second data pass

Resume the mature 92,456-step AdamW42 checkpoint toward step 184,912. Preserve optimizer moments, random state and the exact data cursor. Start the predeclared second-pass cosine schedule at 3e-5 and decay to 3e-6. Original targets, tokenizer, architecture, BF16, batch 128 and complete sequences stay fixed.

| Update | Train min total | Pilot /982 | Confirm /988 | Public /960 | Qualifying gain |
|---|---|---|---|---|---|
| 92,456 | 0 | 615 | 600 | 592 | Parent |
| 102,563 | 16.7 | 613 | 610 | 597 | No |
| 112,738 | 33.3 | 622 | 600 | 588 | No |

The scale-up is still in progress; the table contains only retained completed evaluations.

Each interval uses at most 1,000 measured training seconds. Training minutes exclude loading, checkpoint saving, generation and evaluation. Stop after three consecutive intervals without improved agreement on both fixed held-out subsets and no increase in total execution failures, or after the planned pass or time budget. Public answers do not determine this stopping rule.

Every checkpoint and its exact greedy predictions are saved to Drive before download and bot submission. The local collector verifies downloaded model hashes, original question order, decoded token IDs, CSV/JSONL equality, generation settings and the saved coverage ledger. Intentional second-pass repetitions are counted separately from unique original rows. Ledger consistency is not an independent replay of every row ID.

Frozen training source fdd3fe7. The runtime automatically releases its GPU on completion or failure. These results do not establish first place or the one-T4 180-second inference requirement. Evidence: experiments/selected-scale-v1.

Independent CPU reconstruction matches actual saved row-order hash chains, per-shard counts and full-token totals at updates 102,563, 112,738, including the intentional second-pass repeats. It uses checksum-verified original row IDs and cached lengths without importing the production sampler. Future reconstructed updates are predictions, not evidence of completed training.